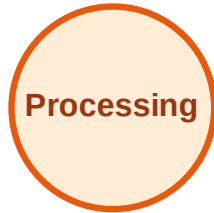


BFS Traversal

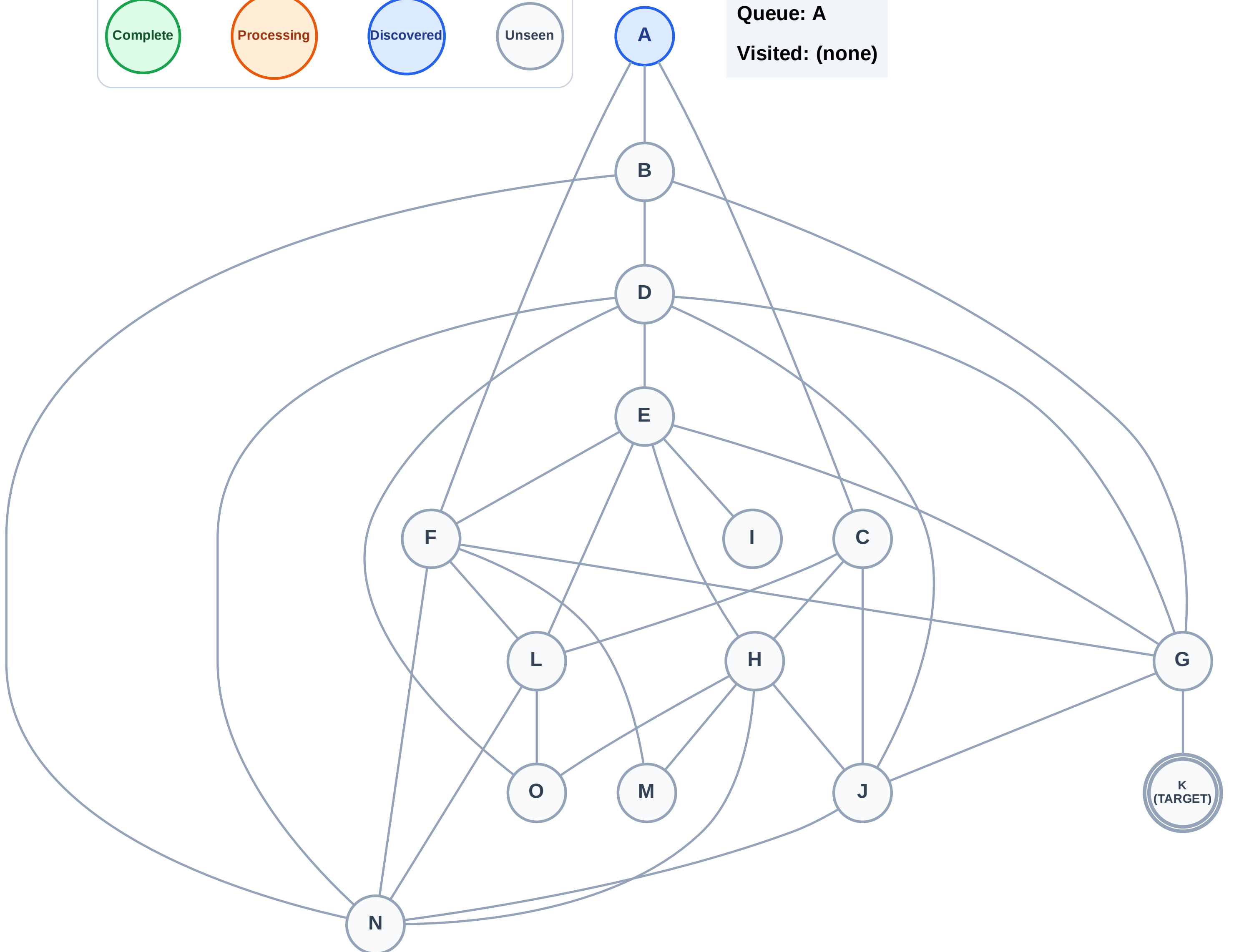
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



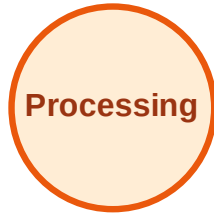
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



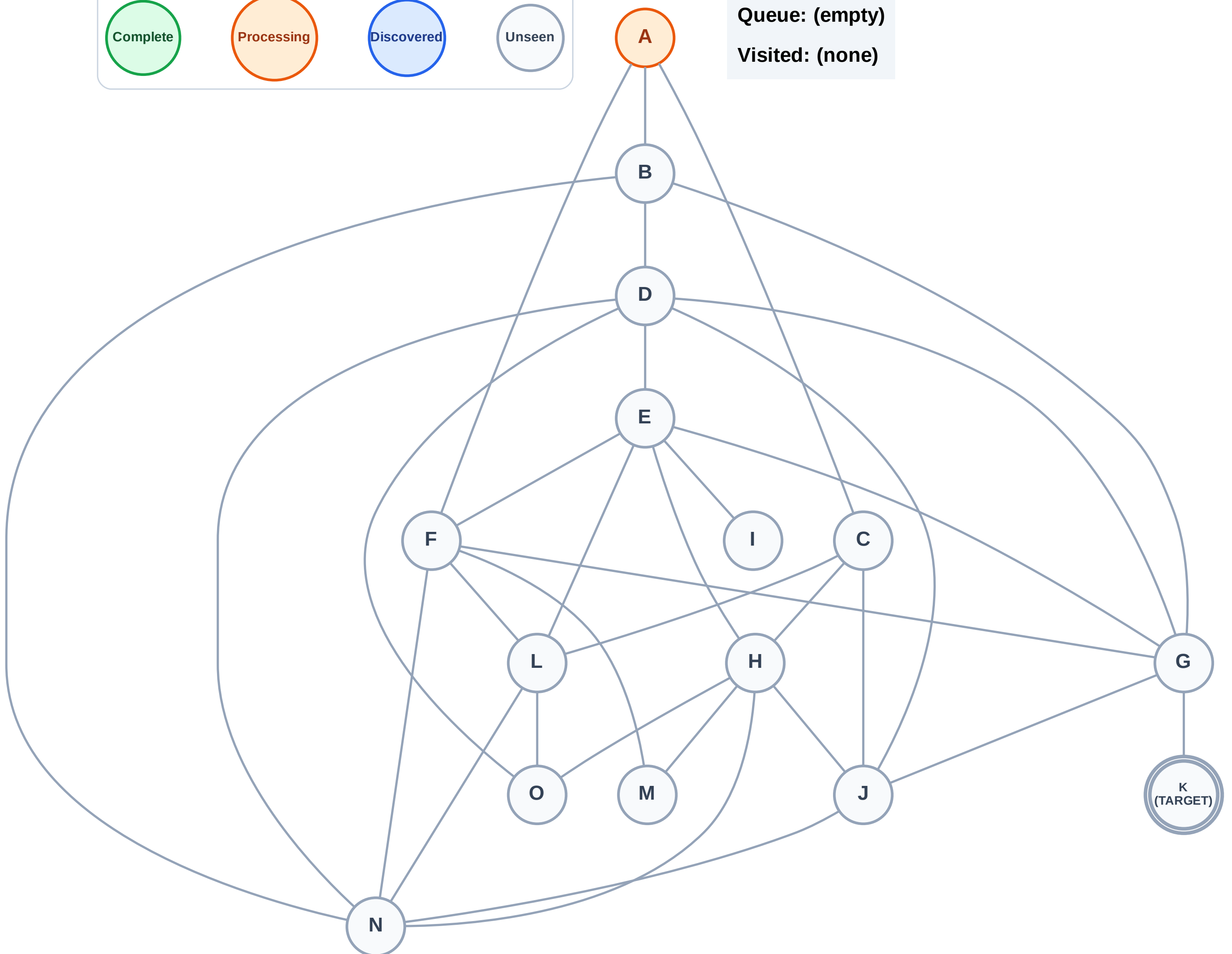
Discovered



Unseen

Queue: (empty)

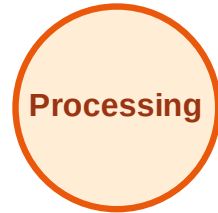
Visited: (none)



BFS Traversal

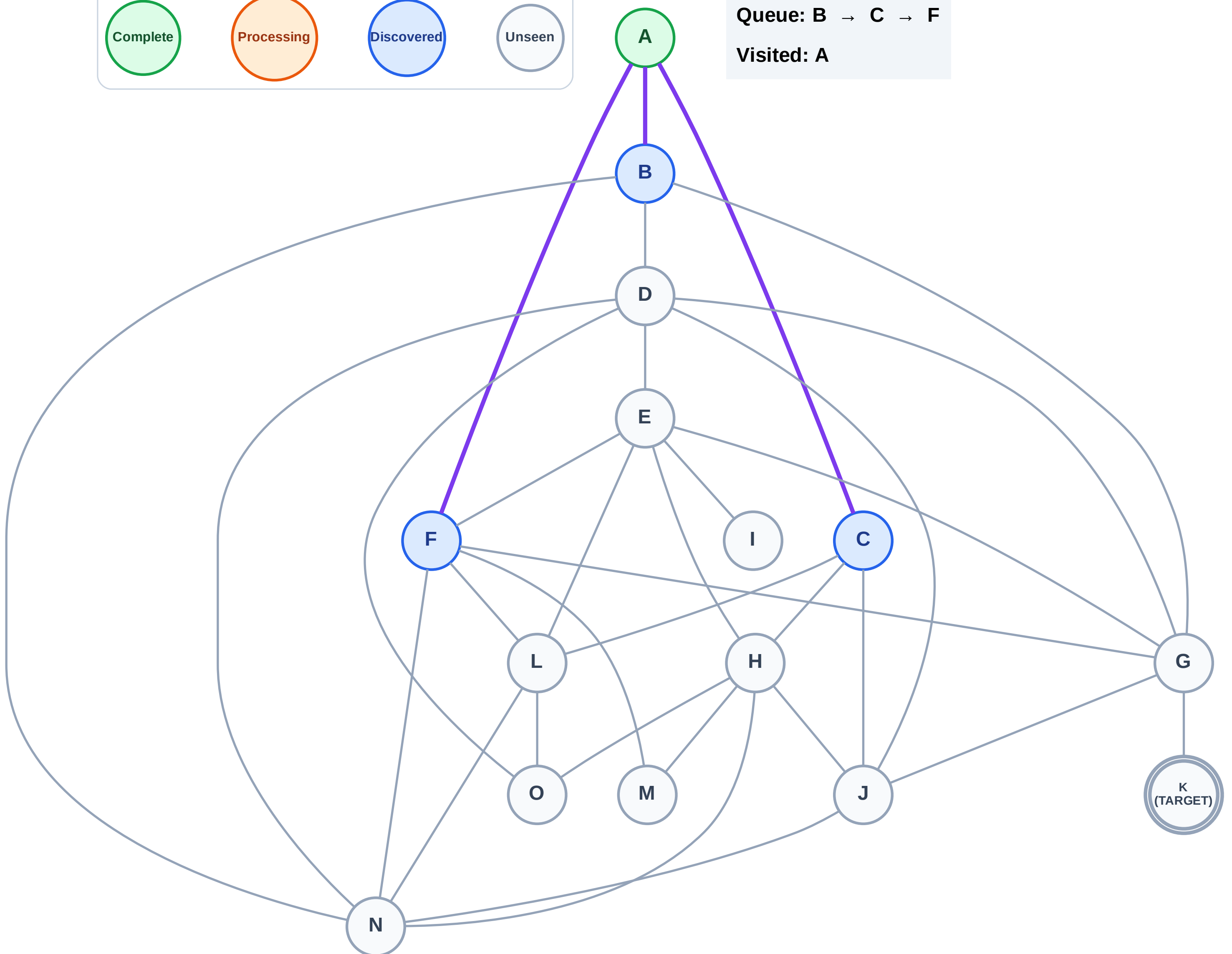
Step 3: Discover B, C, F from A

Legend



Queue: B → C → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

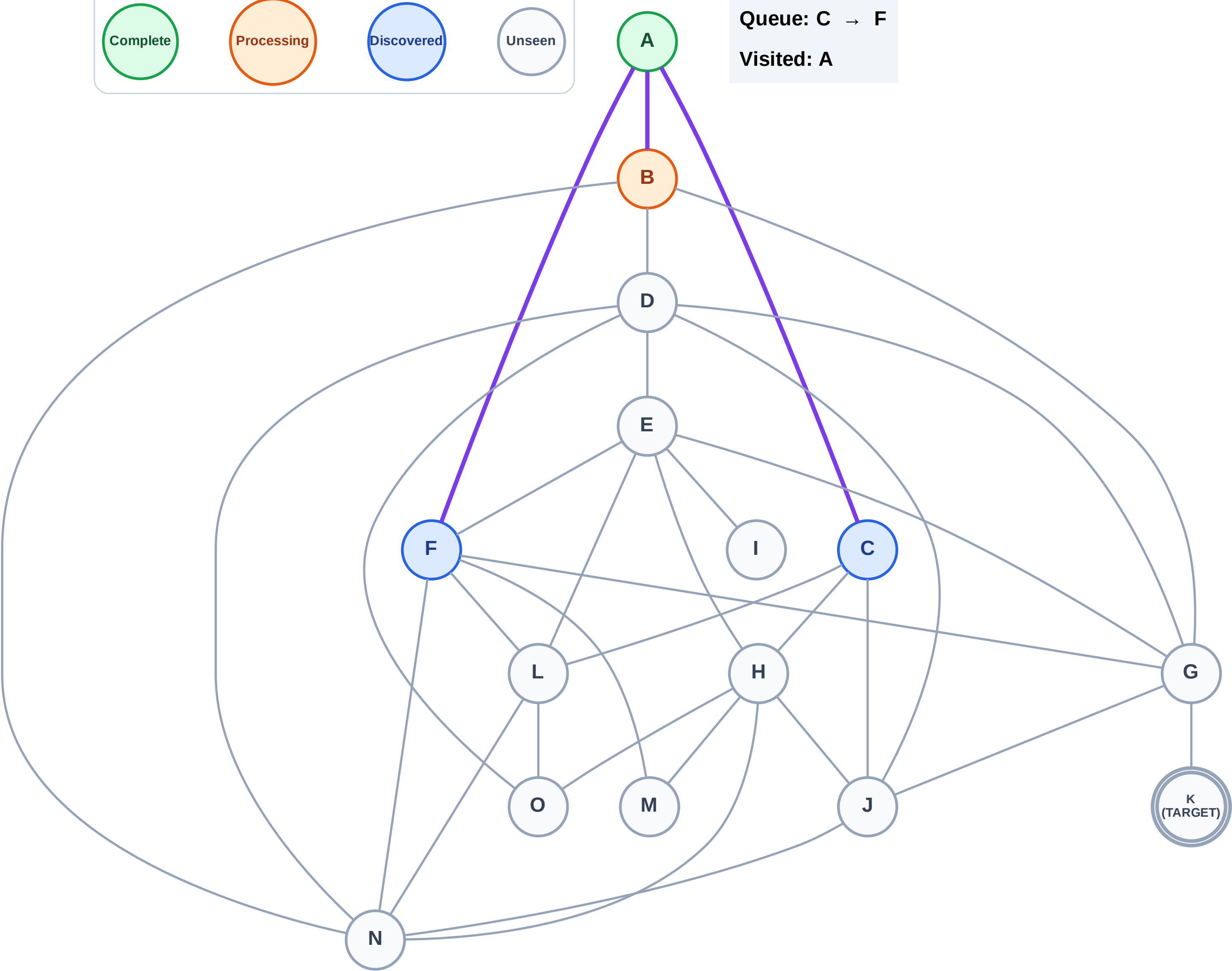
Processing

Discovered

Unseen

Queue: C → F

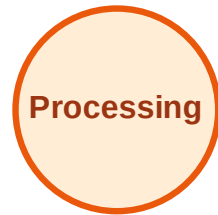
Visited: A



BFS Traversal

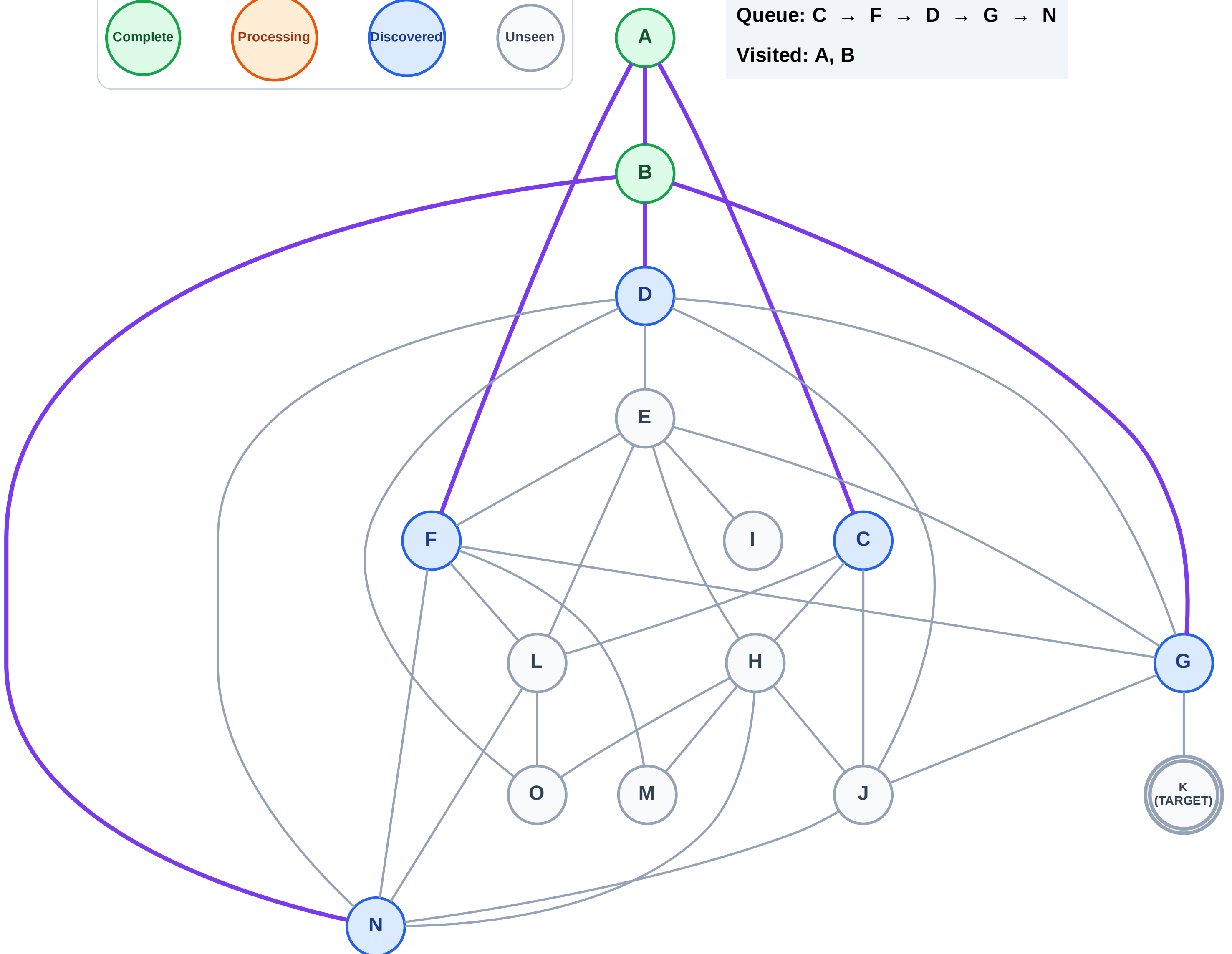
Step 5: Discover D, G, N from B

Legend



Queue: C → F → D → G → N

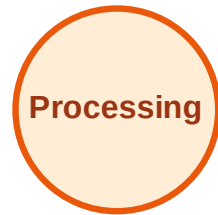
Visited: A, B



BFS Traversal

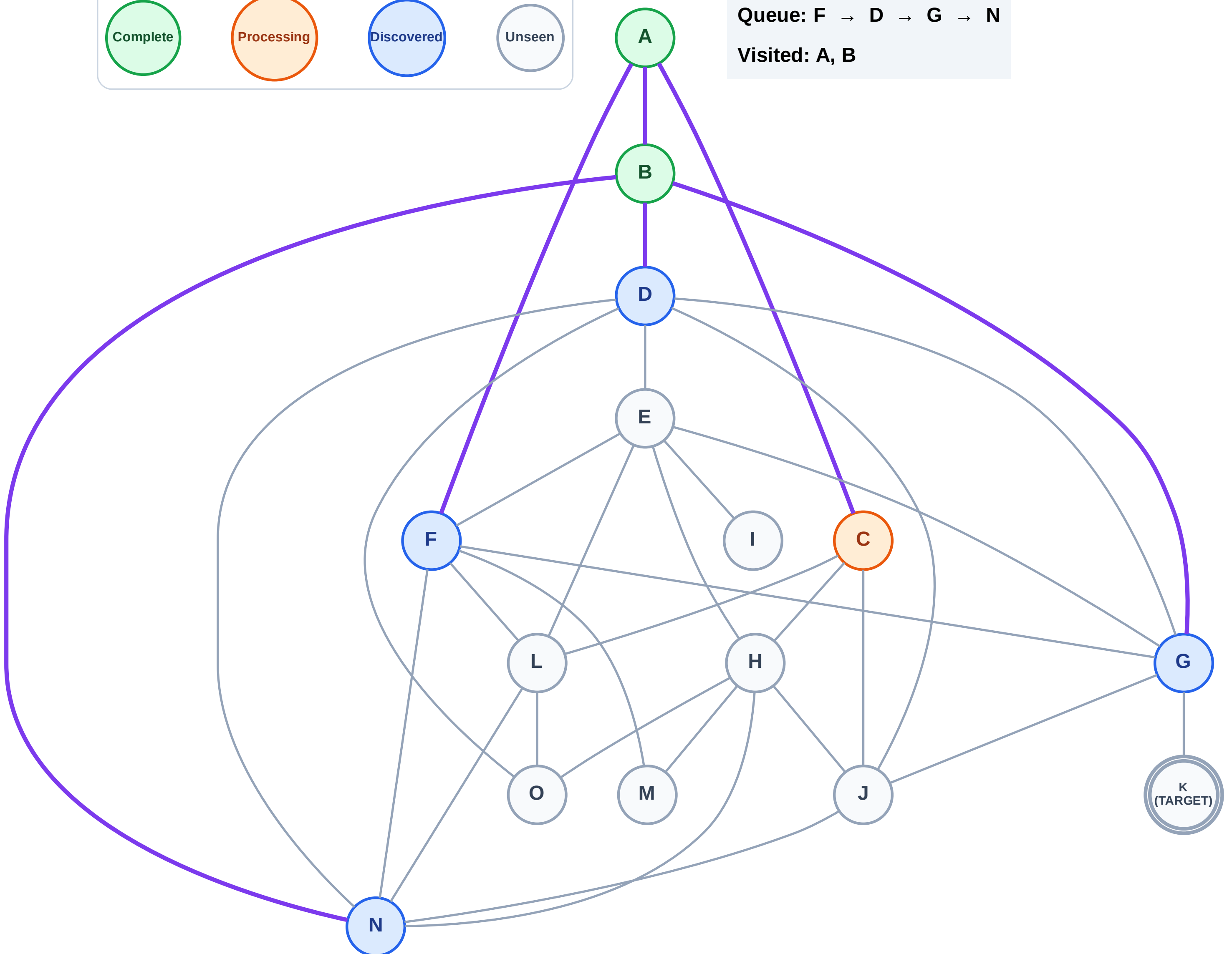
Step 6: Process node C

Legend



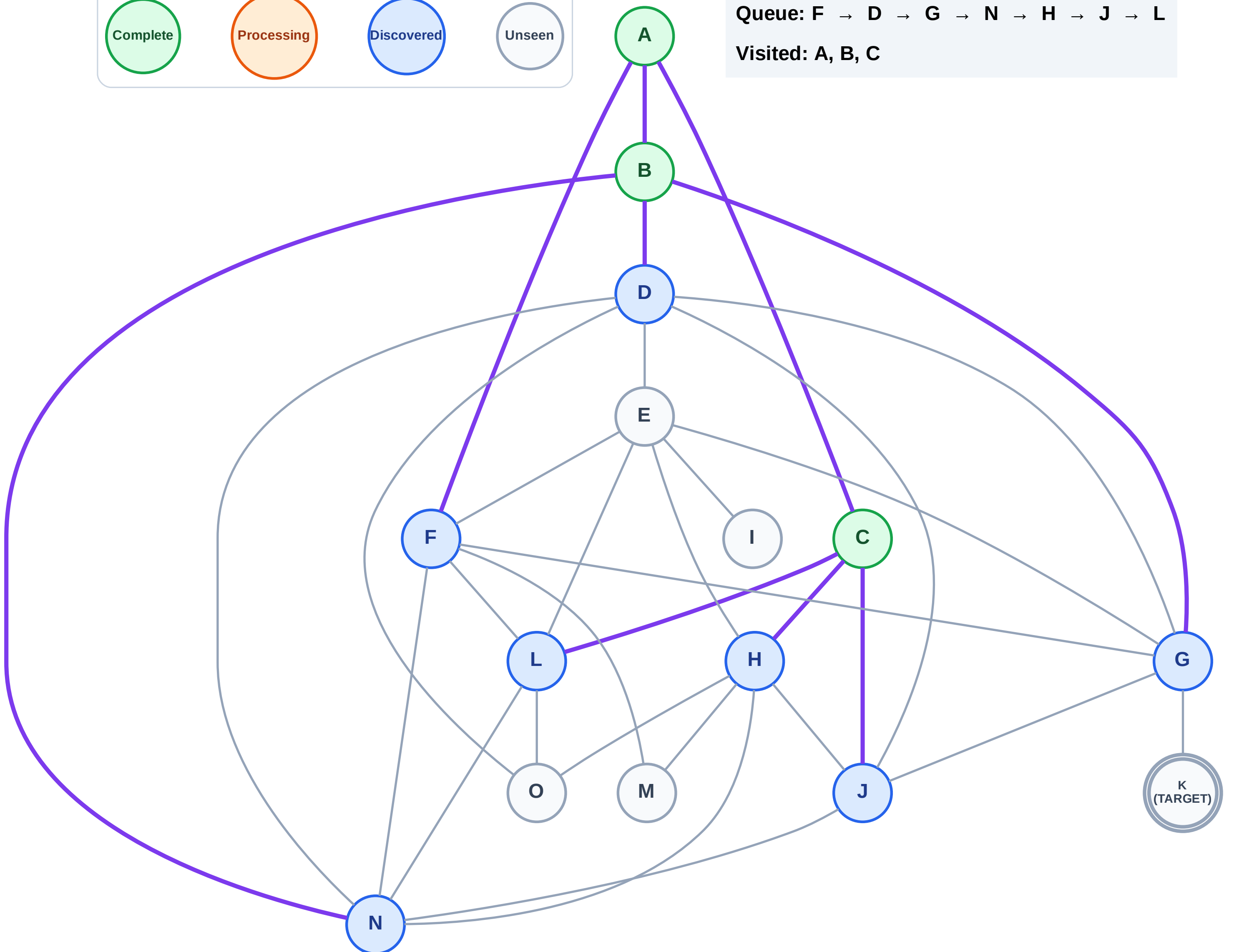
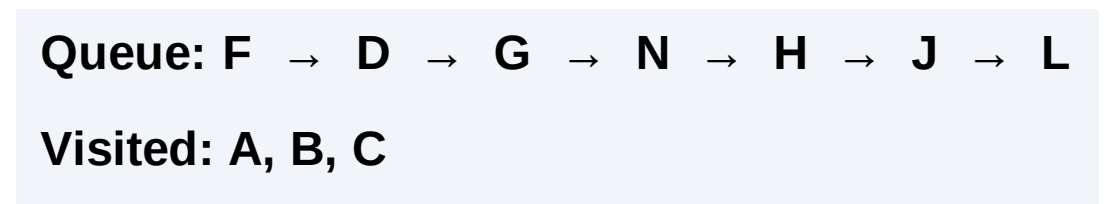
Queue: F → D → G → N

Visited: A, B



Step 7: Discover H, J, L from C

Step 7: Discover H, J, L from C



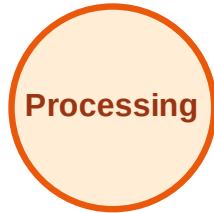
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



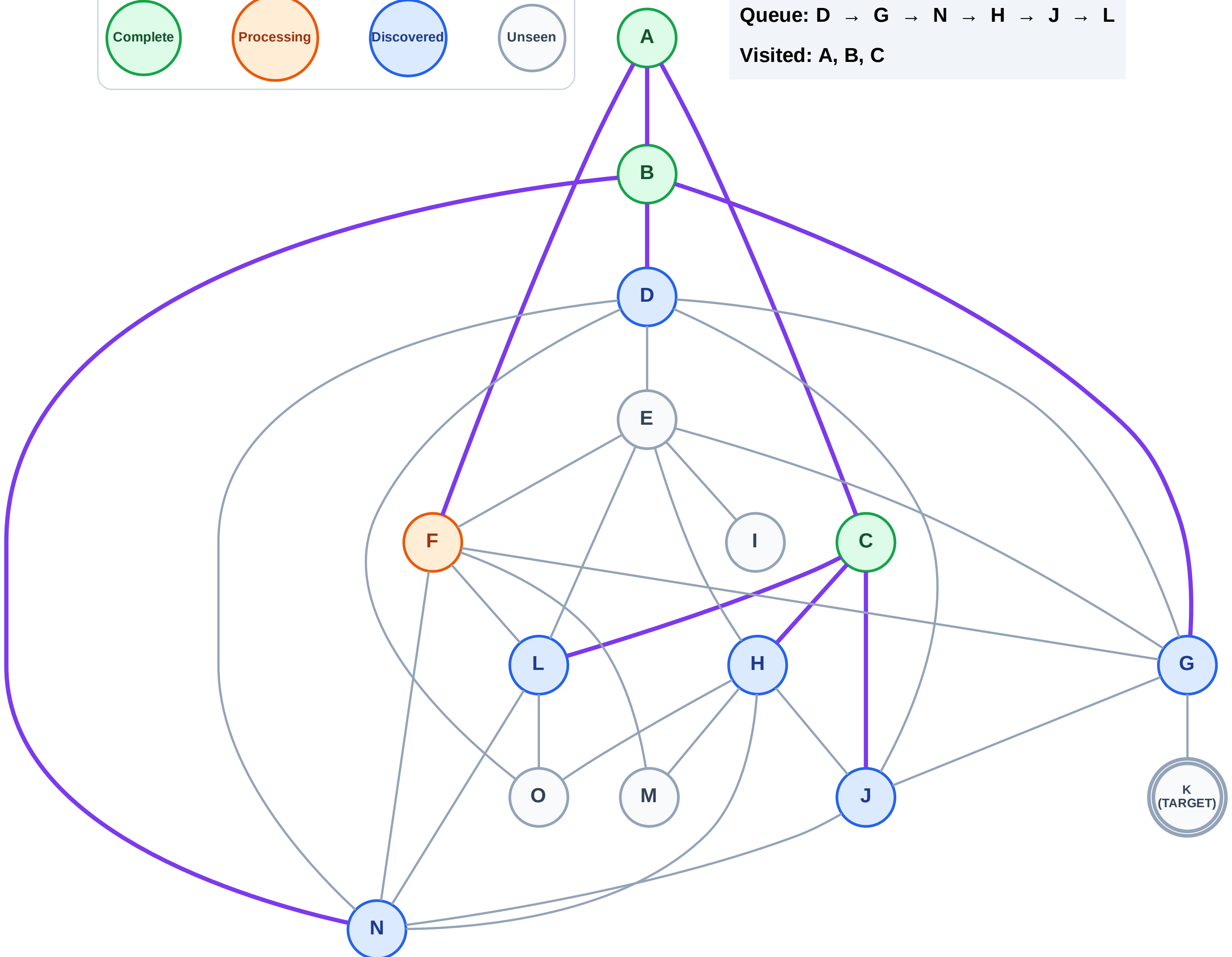
Discovered



Unseen

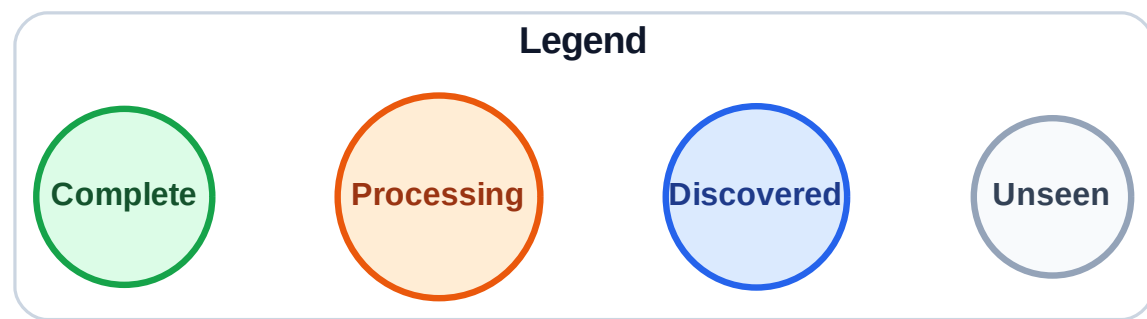
Queue: D → G → N → H → J → L

Visited: A, B, C



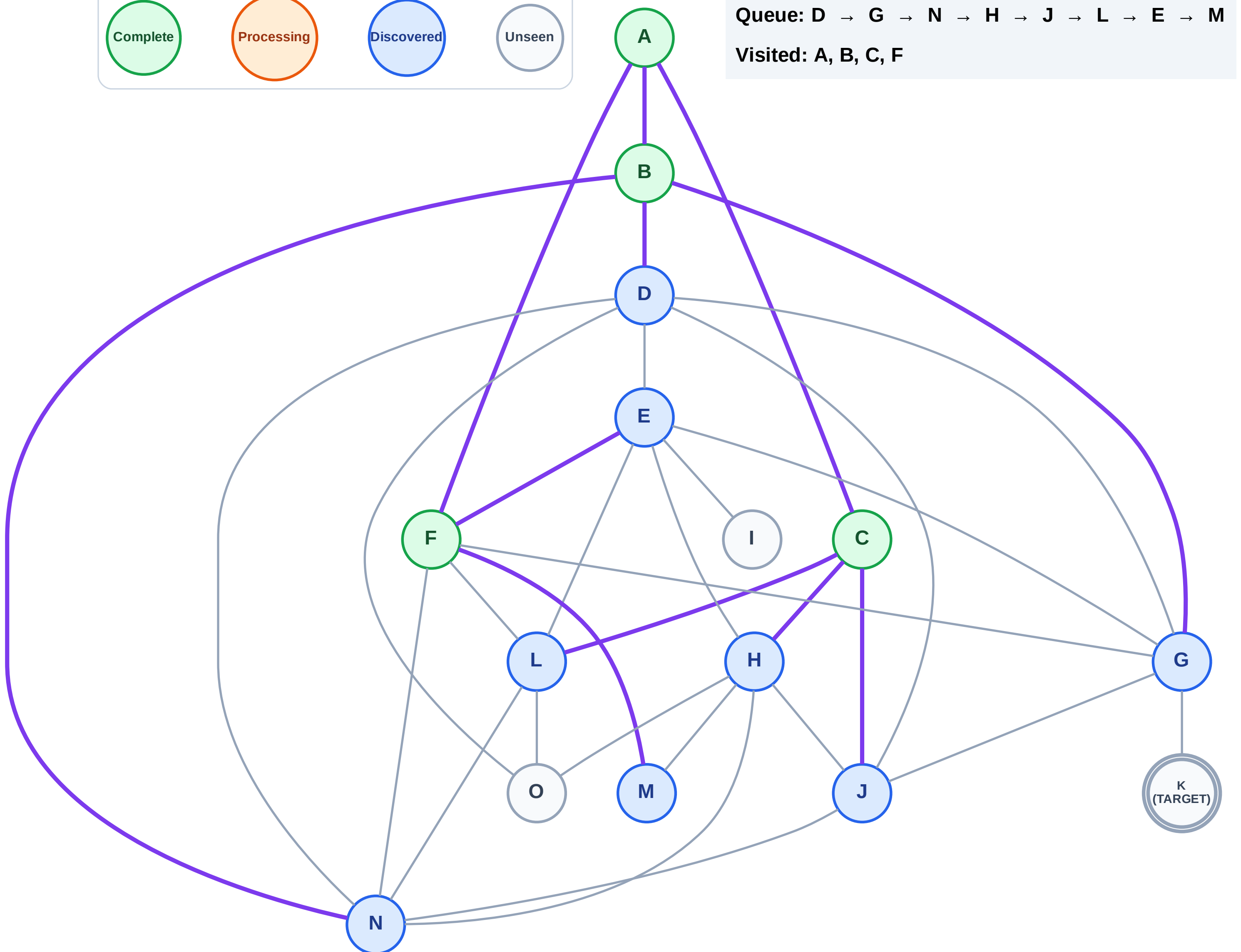
BFS Traversal

Step 9: Discover E, M from F



Queue: D → G → N → H → J → L → E → M

Visited: A, B, C, F



BFS Traversal

Step 10: Process node D

Legend

Complete

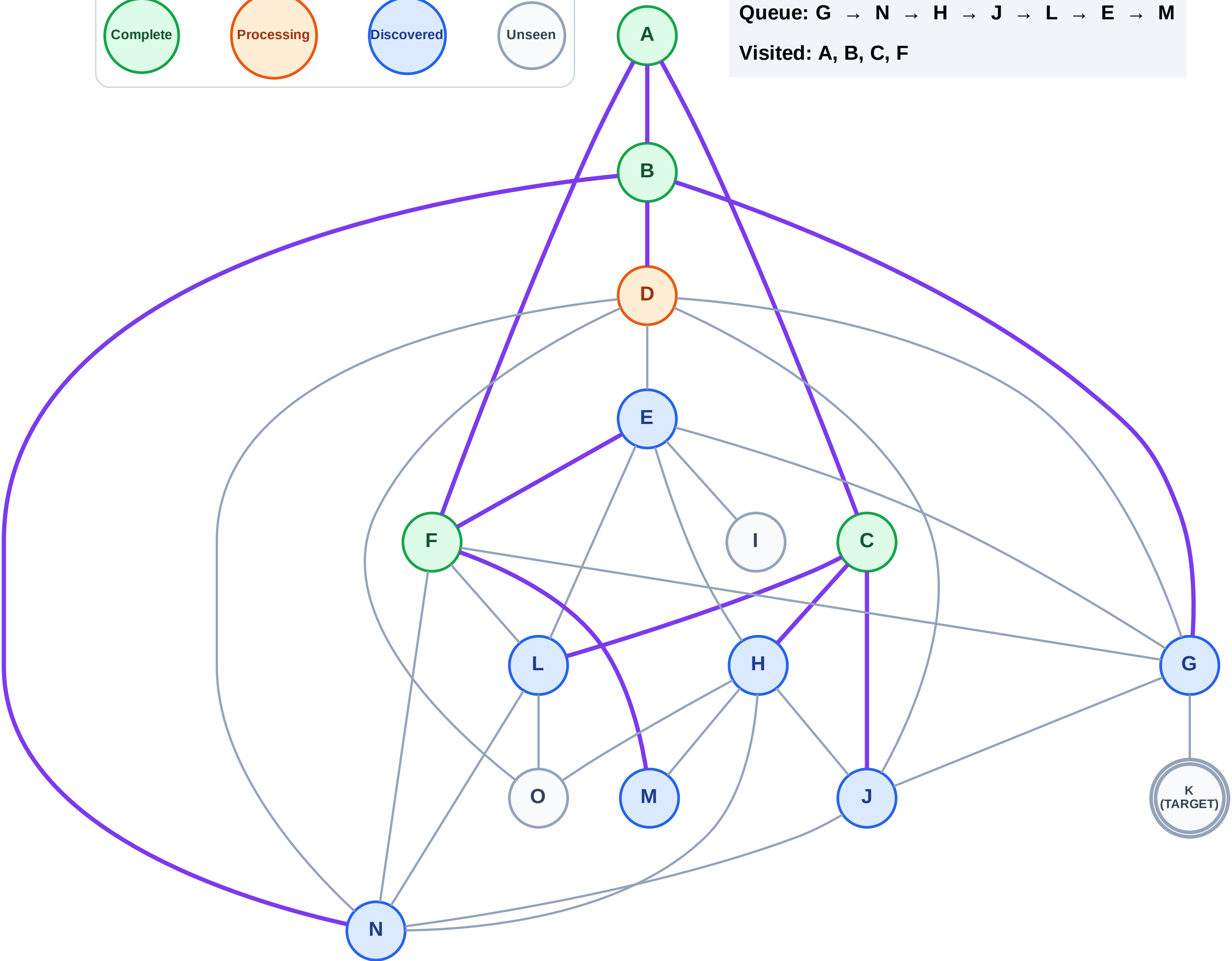
Processing

Discovered

Unseen

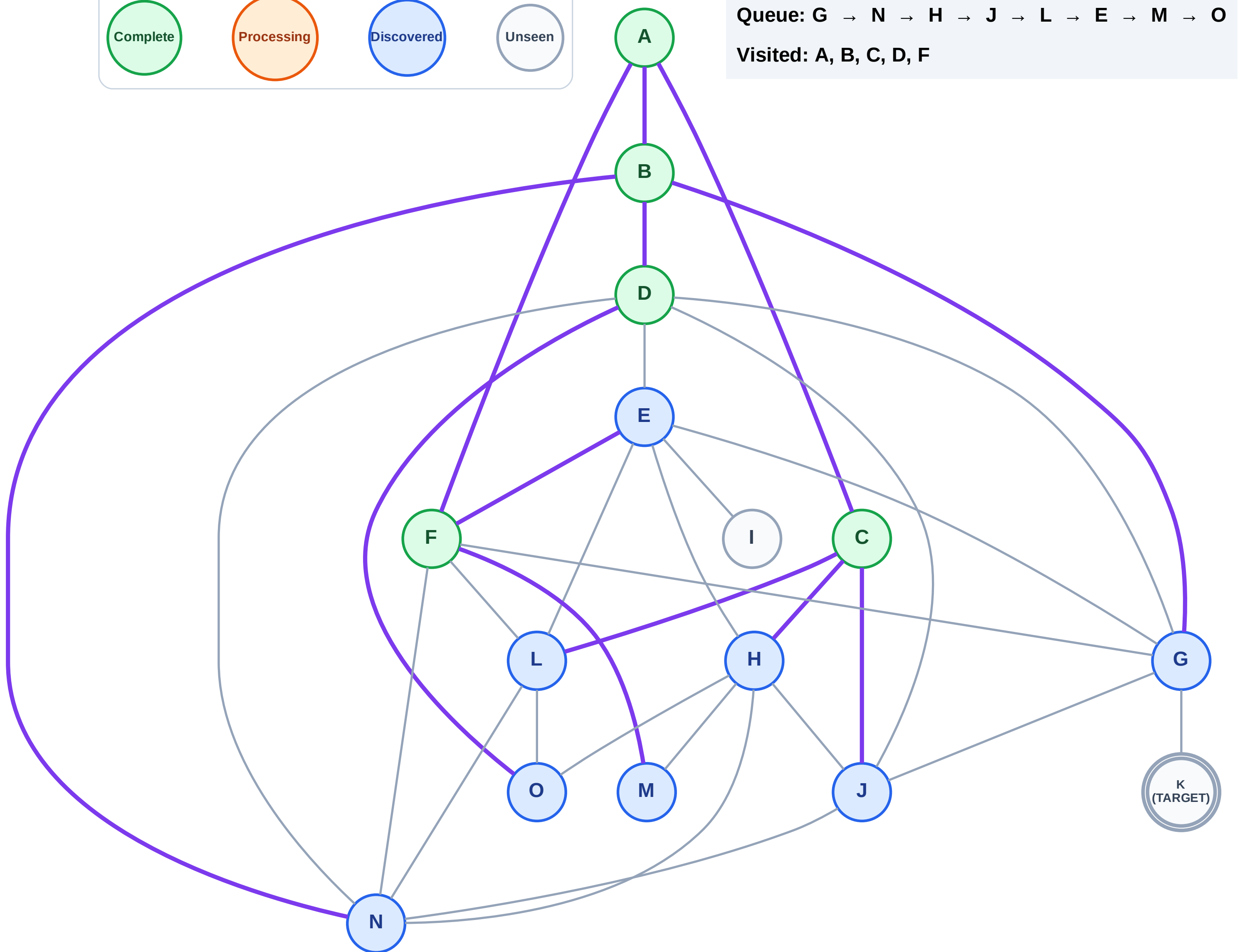
Queue: G → N → H → J → L → E → M

Visited: A, B, C, F



Step 11: Discover O from D

Step 11: Discover O from D

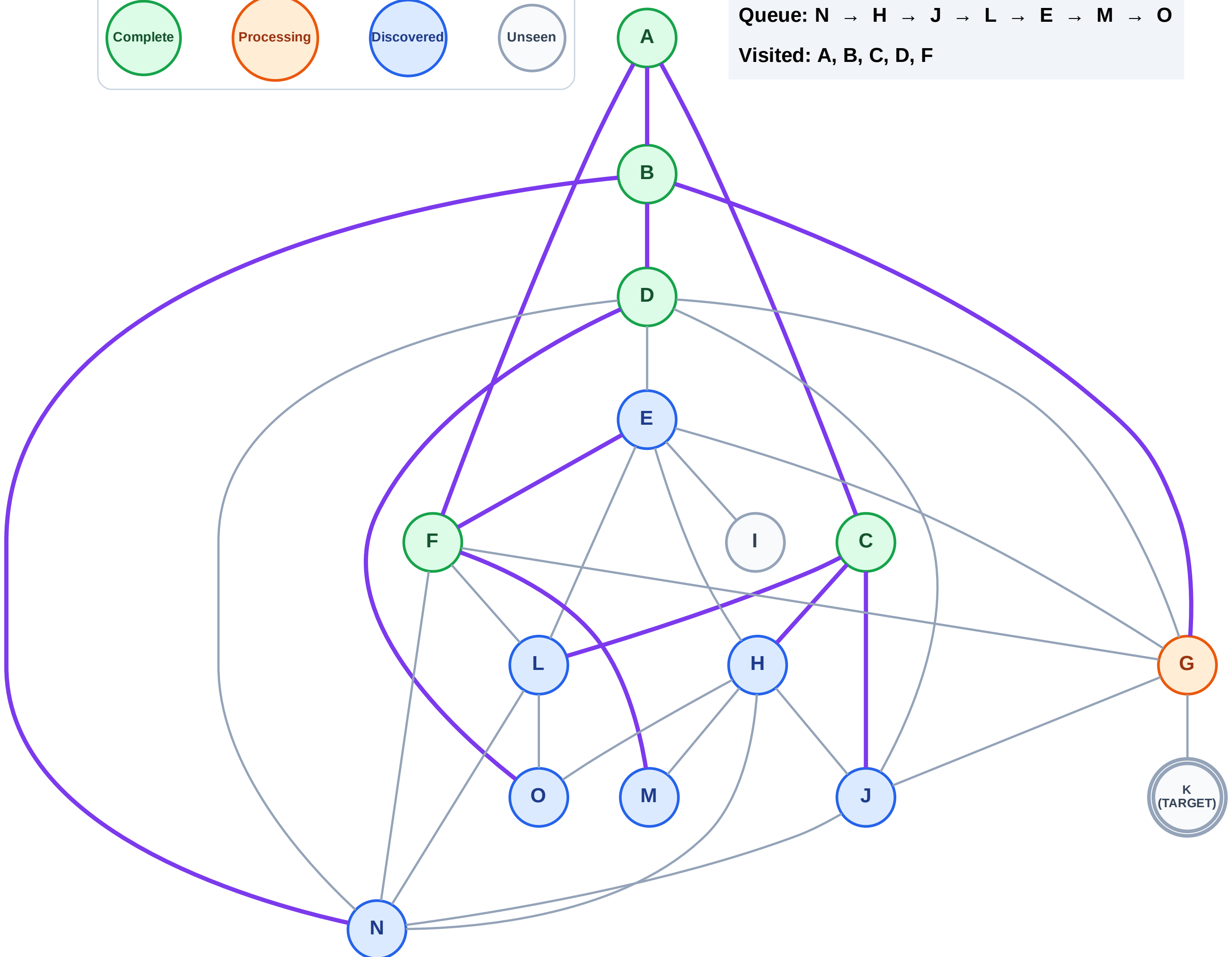


Step 12: Process node G

Legend

Unseen

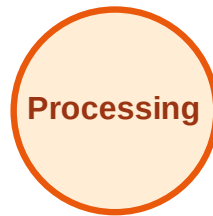
Visited: A, B, C, D, F



BFS Traversal

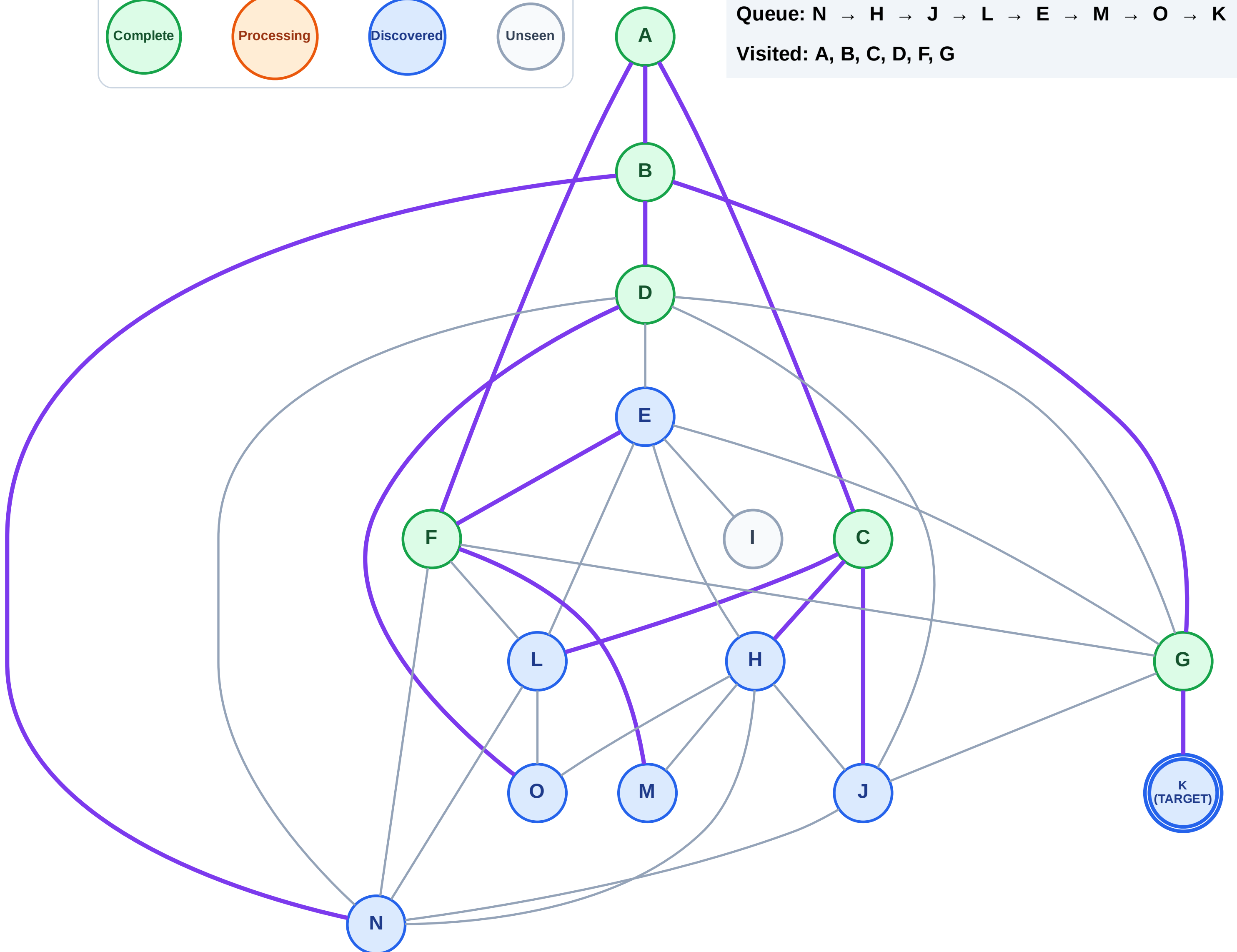
Step 13: Discover K from G

Legend



Queue: N → H → J → L → E → M → O → K

Visited: A, B, C, D, F, G



BFS Traversal

Step 14: Process node N

Legend

Complete

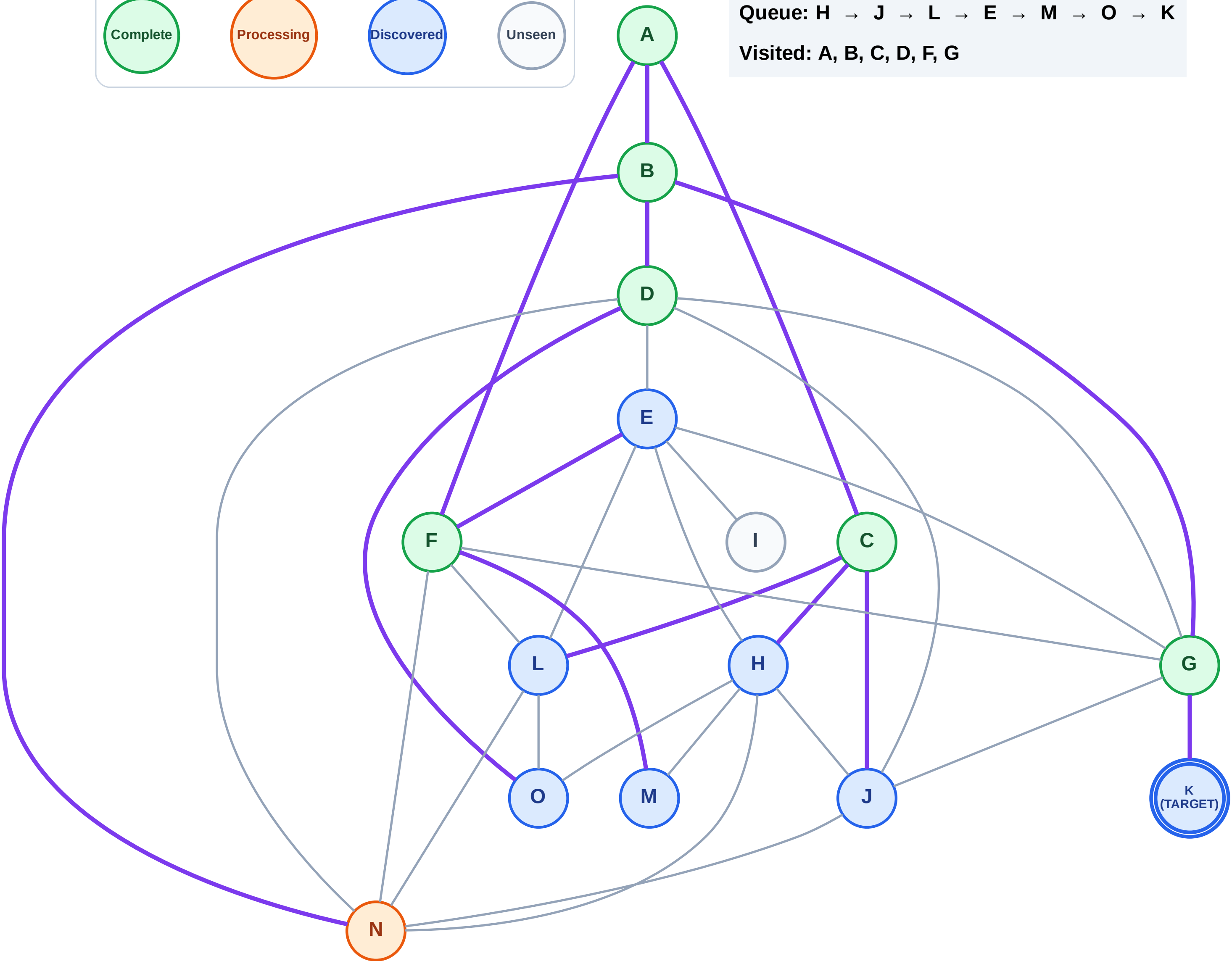
Processing

Discovered

Unseen

Queue: H → J → L → E → M → O → K

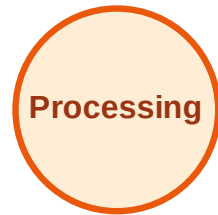
Visited: A, B, C, D, F, G



BFS Traversal

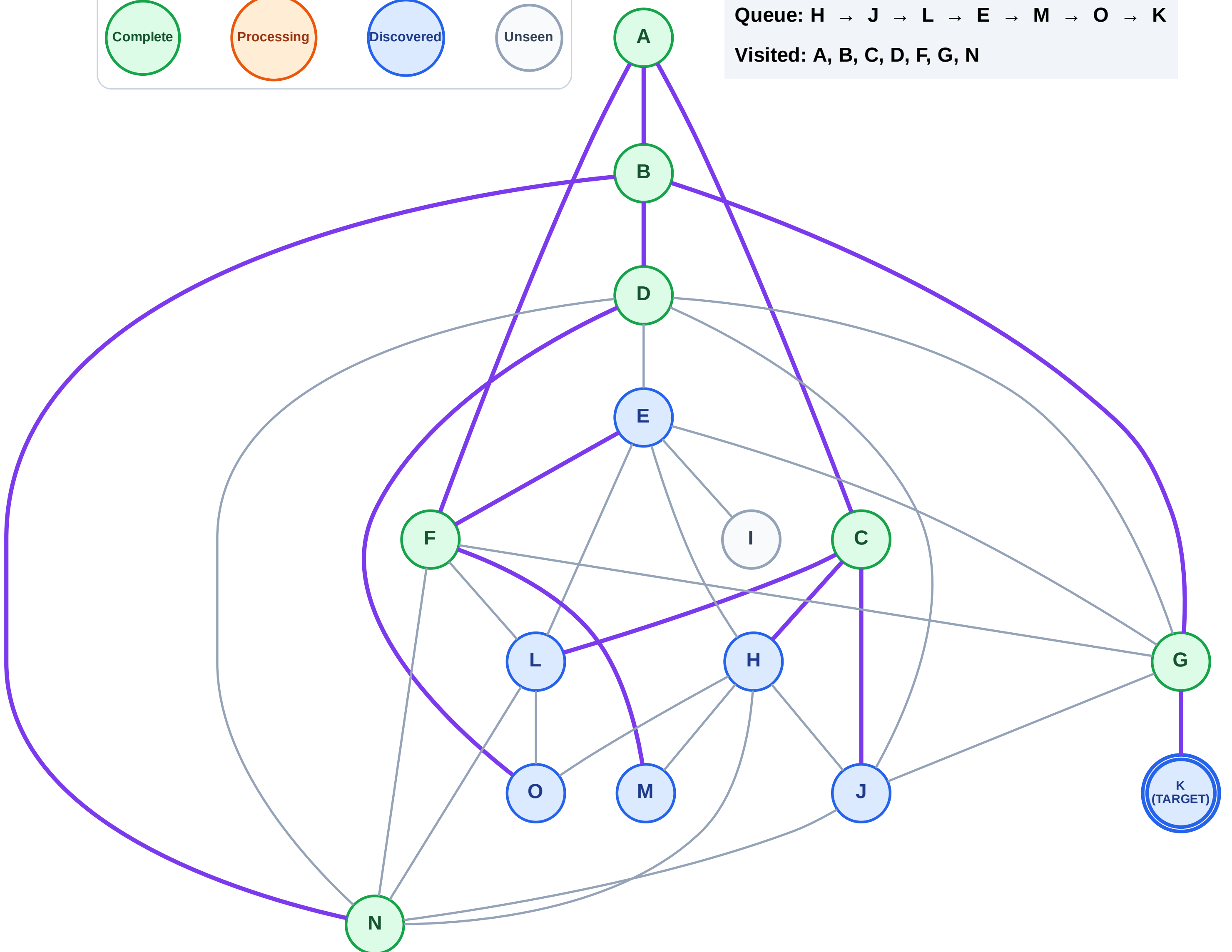
Step 15: No new neighbors from N

Legend



Queue: H → J → L → E → M → O → K

Visited: A, B, C, D, F, G, N



BFS Traversal

Step 16: Process node H

Legend

Complete

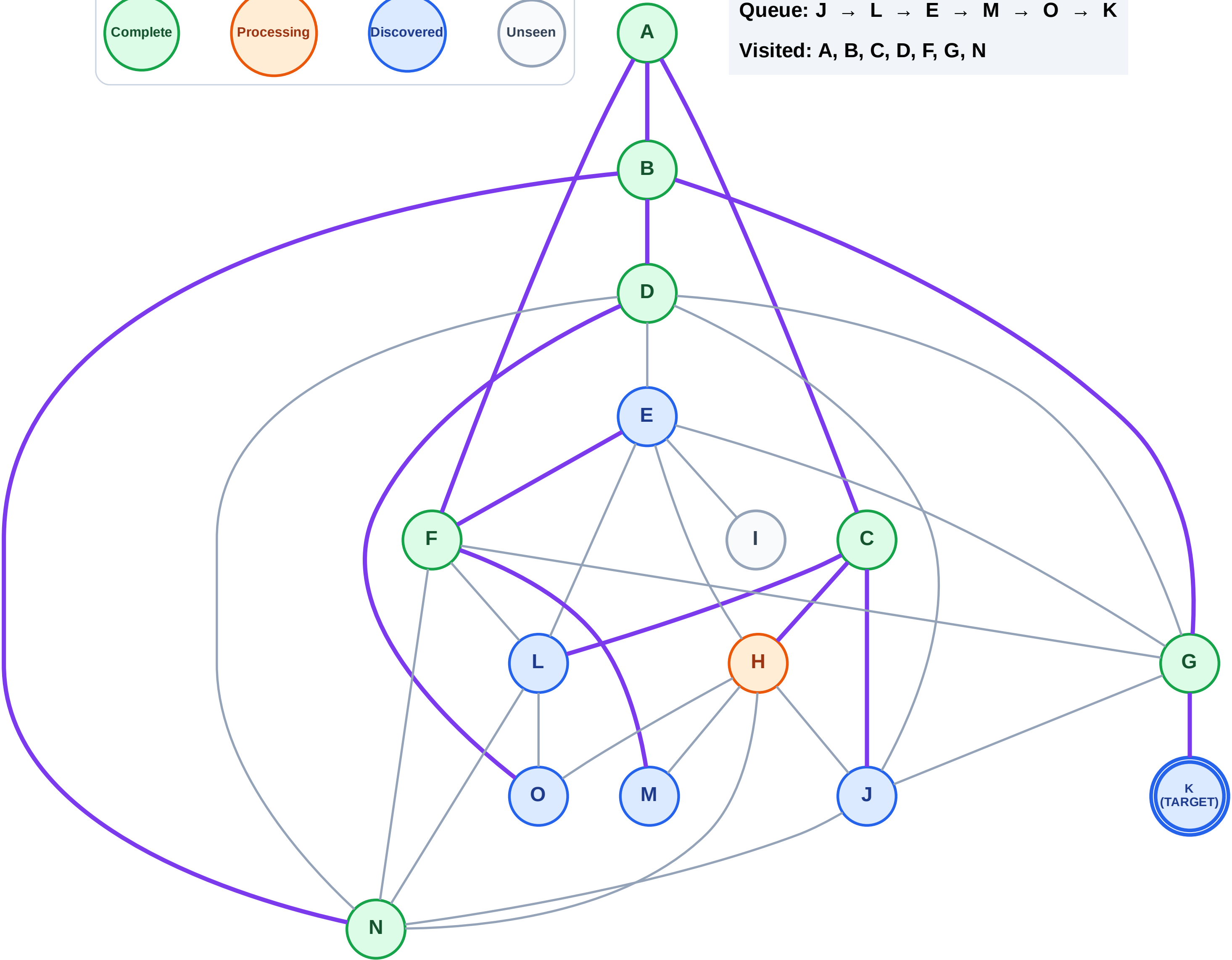
Processing

Discovered

Unseen

Queue: J → L → E → M → O → K

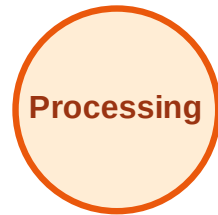
Visited: A, B, C, D, F, G, N



BFS Traversal

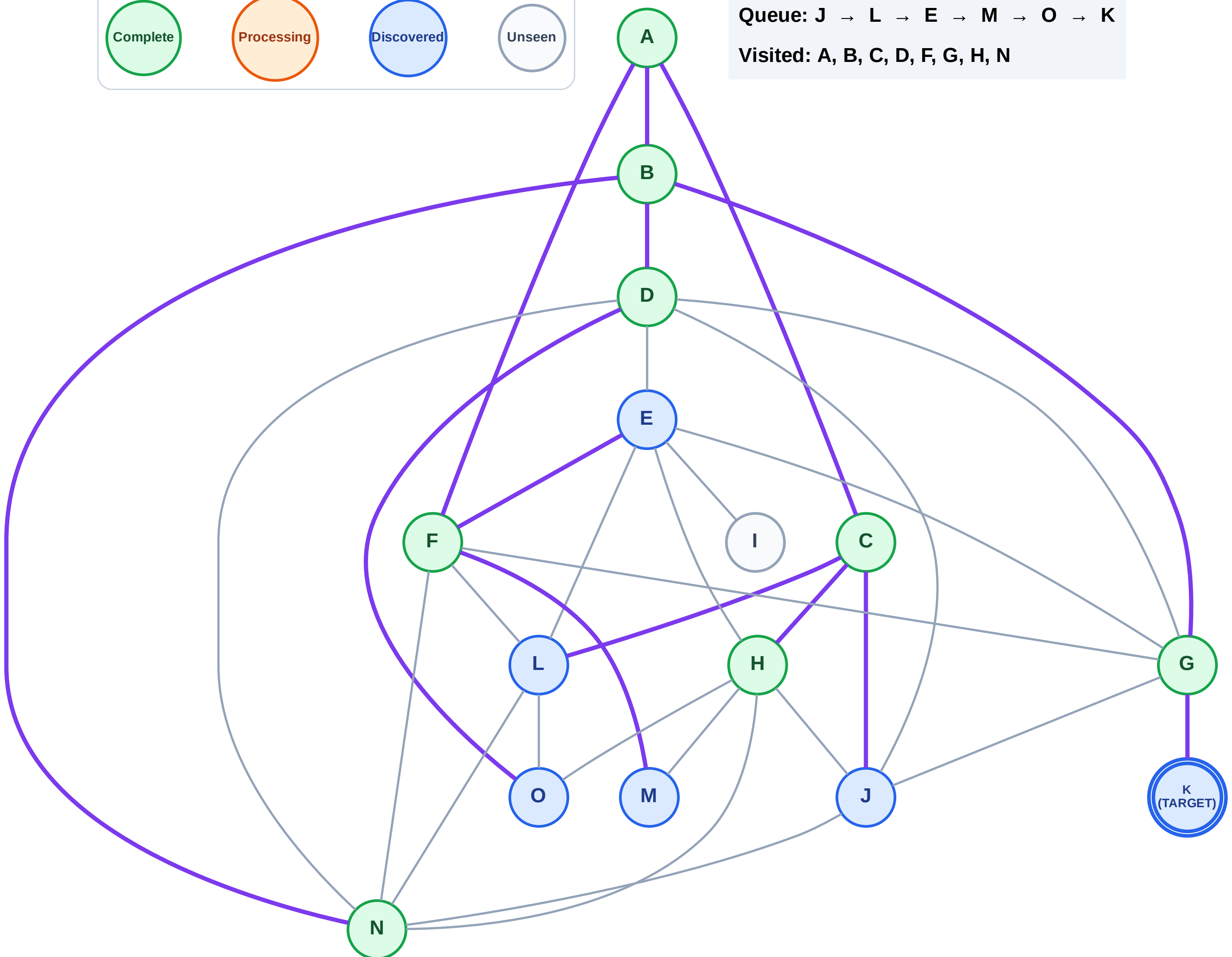
Step 17: No new neighbors from H

Legend



Queue: J → L → E → M → O → K

Visited: A, B, C, D, F, G, H, N



BFS Traversal

Step 18: Process node J

Legend

Complete

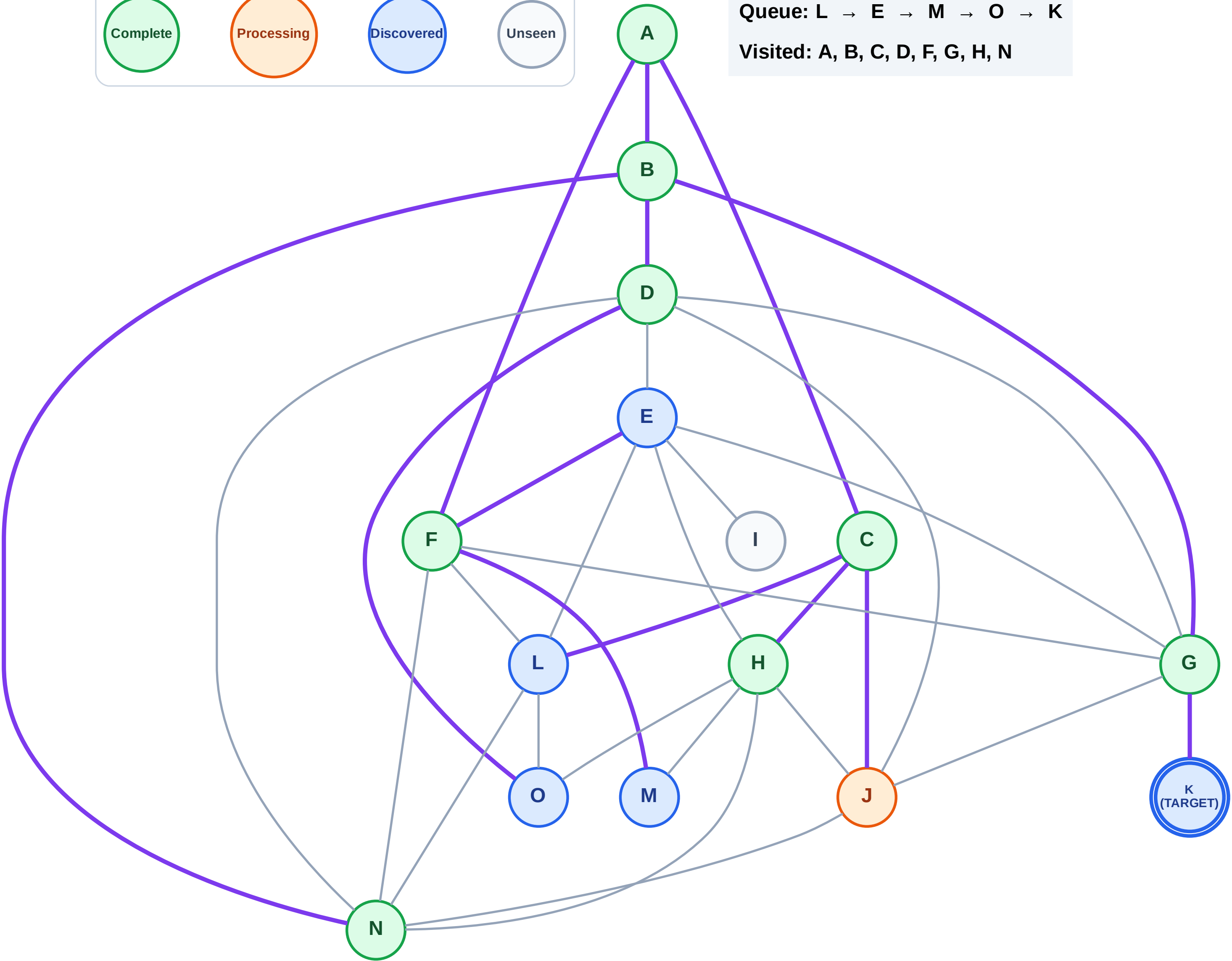
Processing

Discovered

Unseen

Queue: L → E → M → O → K

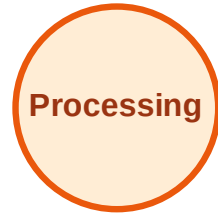
Visited: A, B, C, D, F, G, H, N



BFS Traversal

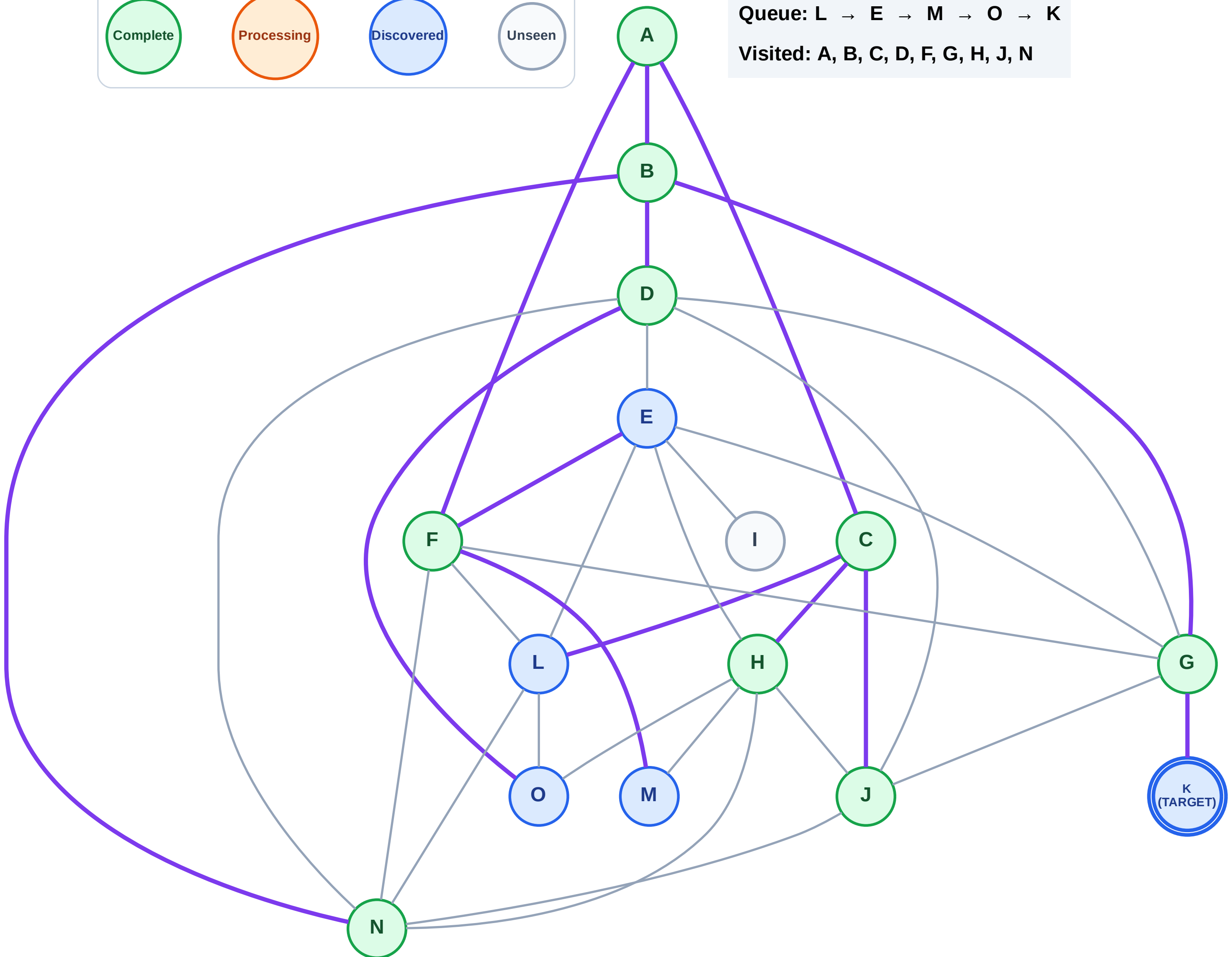
Step 19: No new neighbors from J

Legend



Queue: L → E → M → O → K

Visited: A, B, C, D, F, G, H, J, N



BFS Traversal

Step 20: Process node L

Legend

Complete

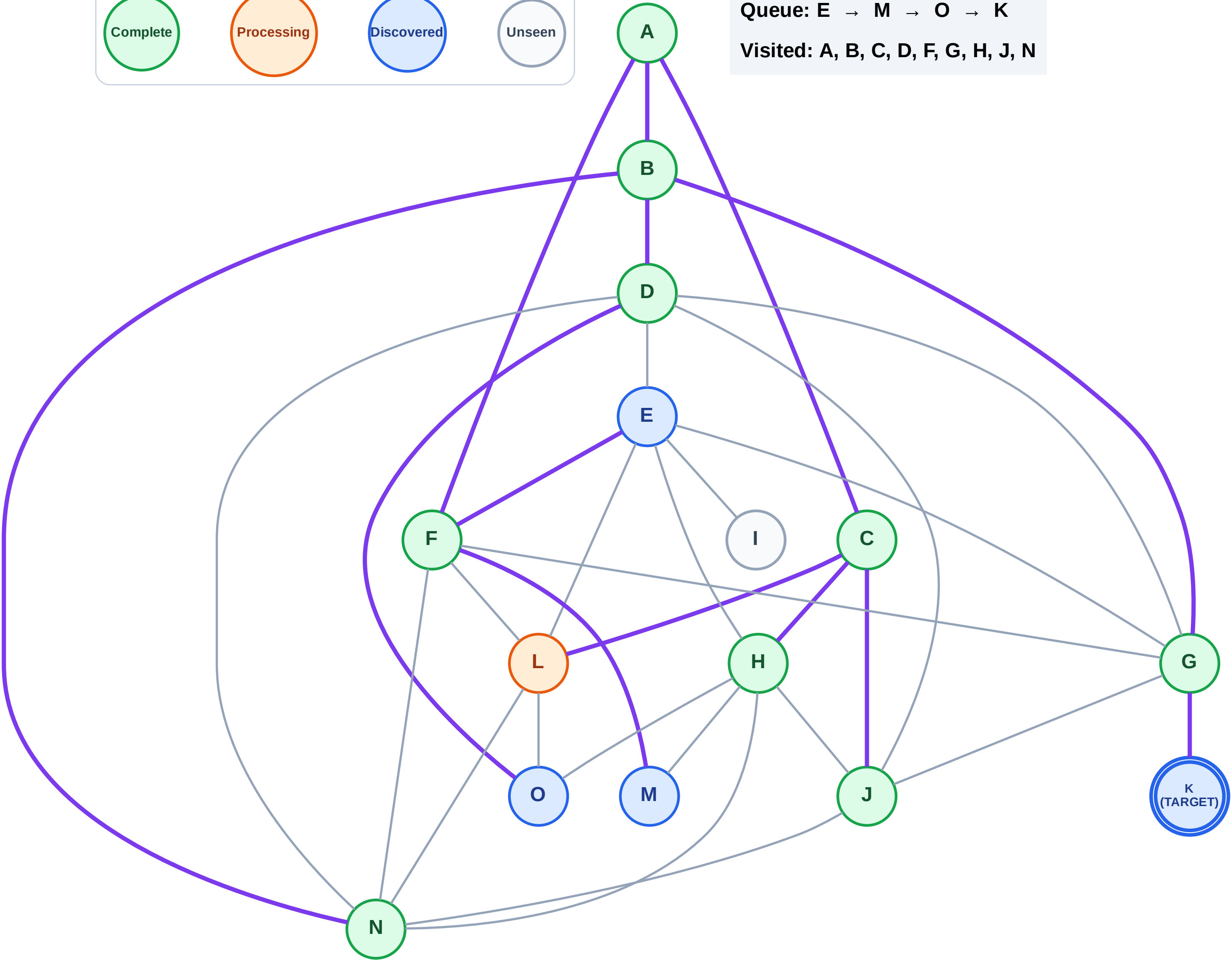
Processing

Discovered

Unseen

Queue: E → M → O → K

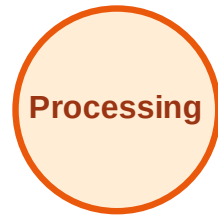
Visited: A, B, C, D, F, G, H, J, N



BFS Traversal

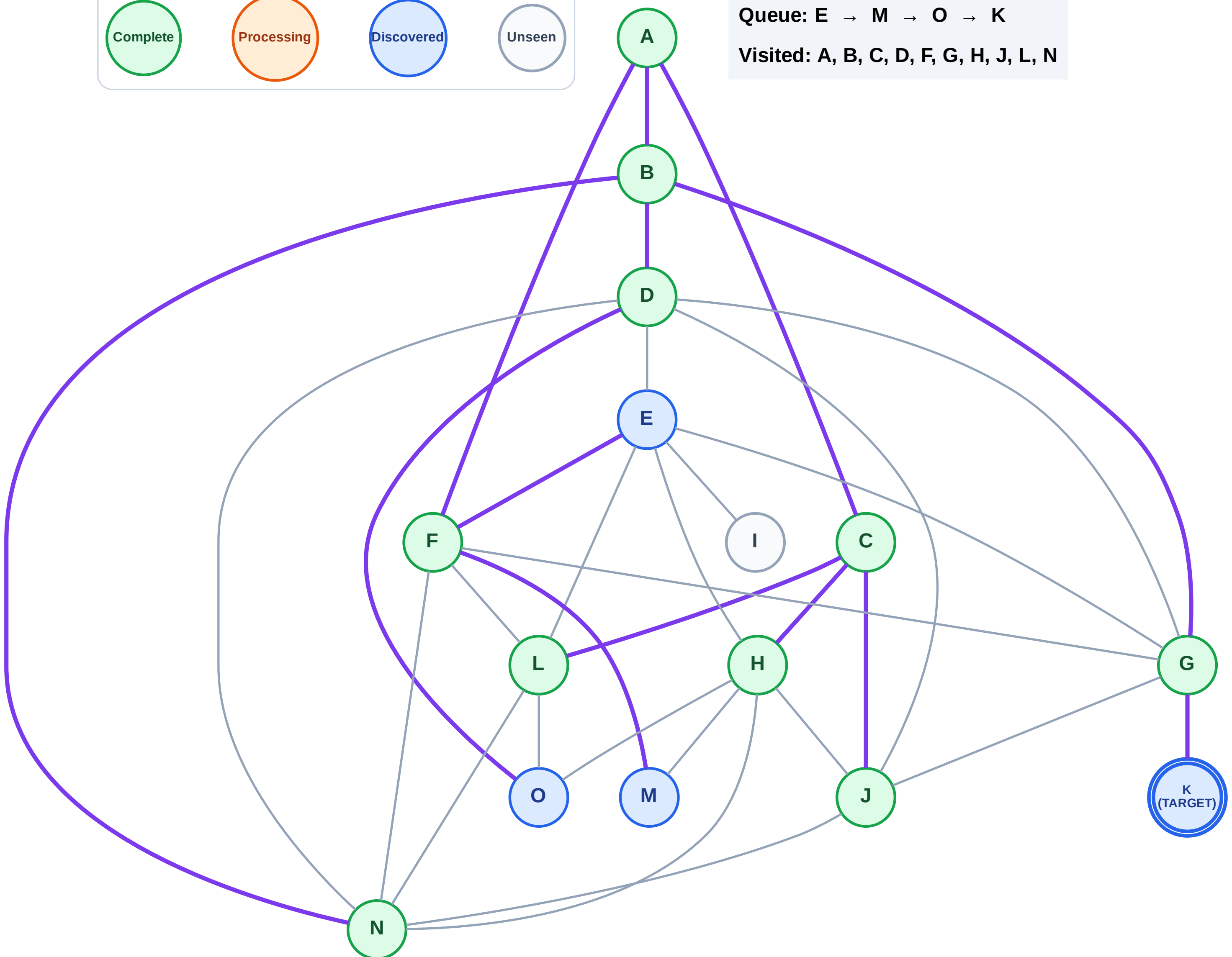
Step 21: No new neighbors from L

Legend



Queue: E → M → O → K

Visited: A, B, C, D, F, G, H, J, L, N

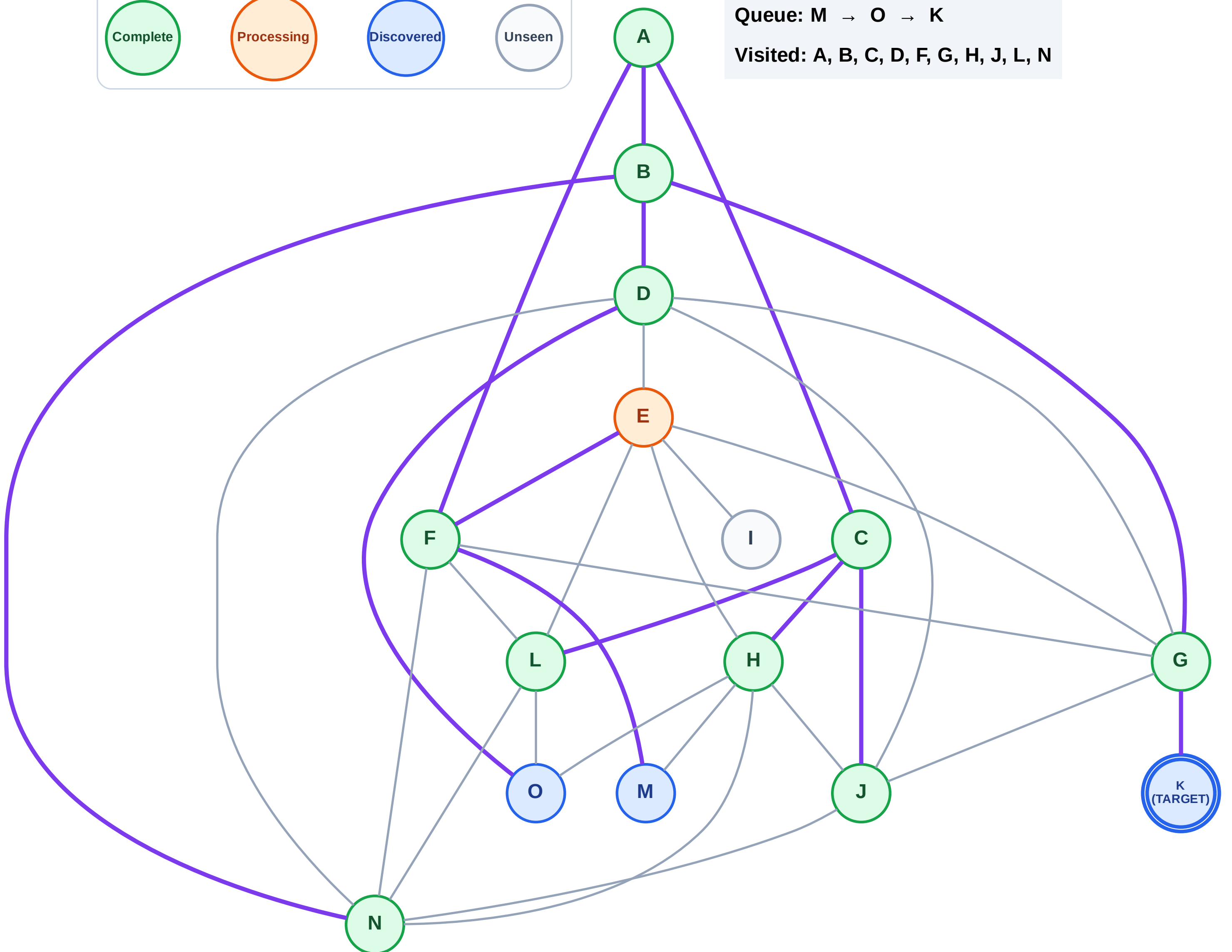


Step 22: Process node E

Legend

Unseen

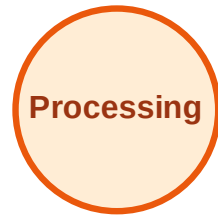
Visited: A, B, C, D, F, G, H, J, L, N



BFS Traversal

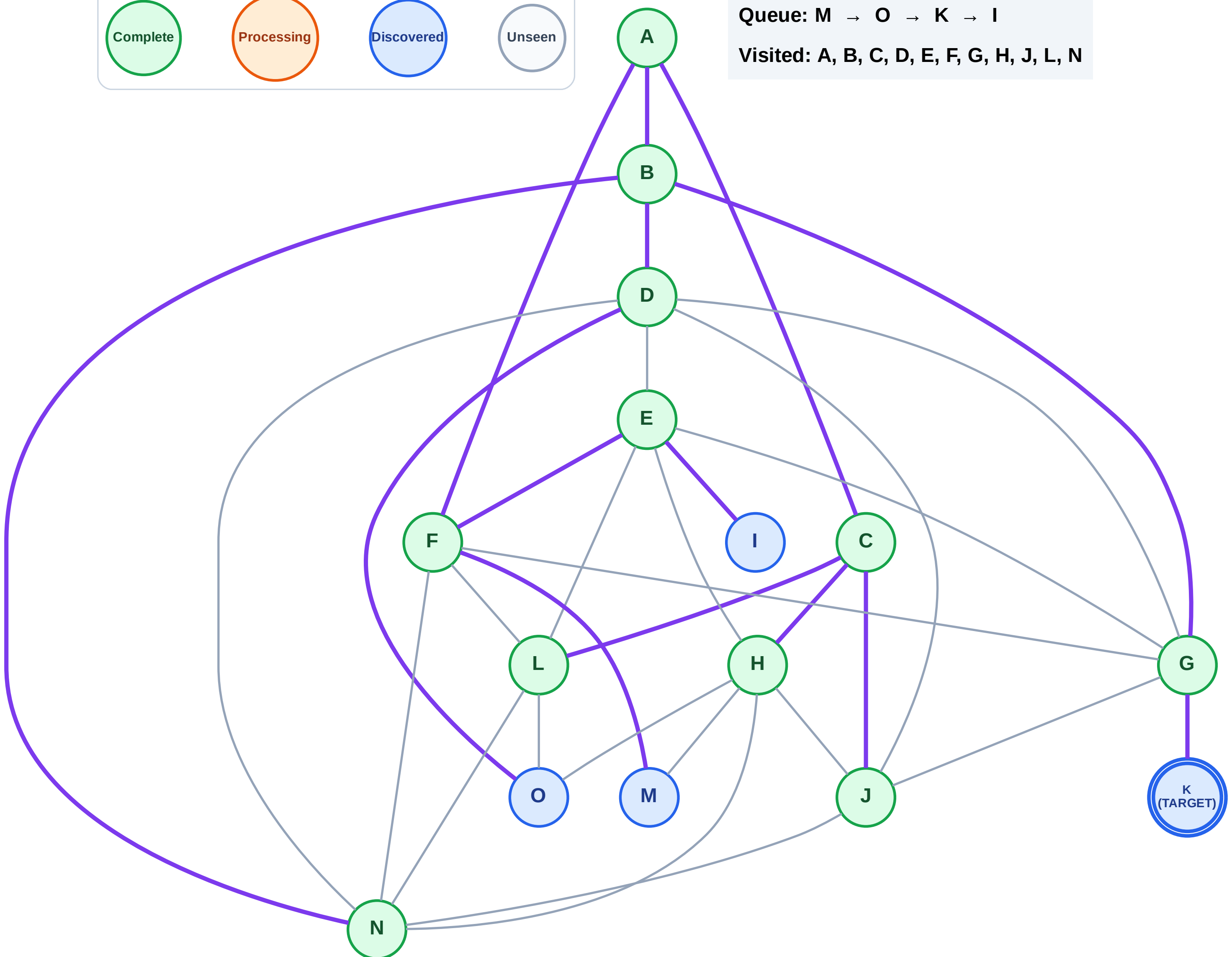
Step 23: Discover I from E

Legend



Queue: M → O → K → I

Visited: A, B, C, D, E, F, G, H, J, L, N



BFS Traversal

Step 24: Process node M

Legend

Complete

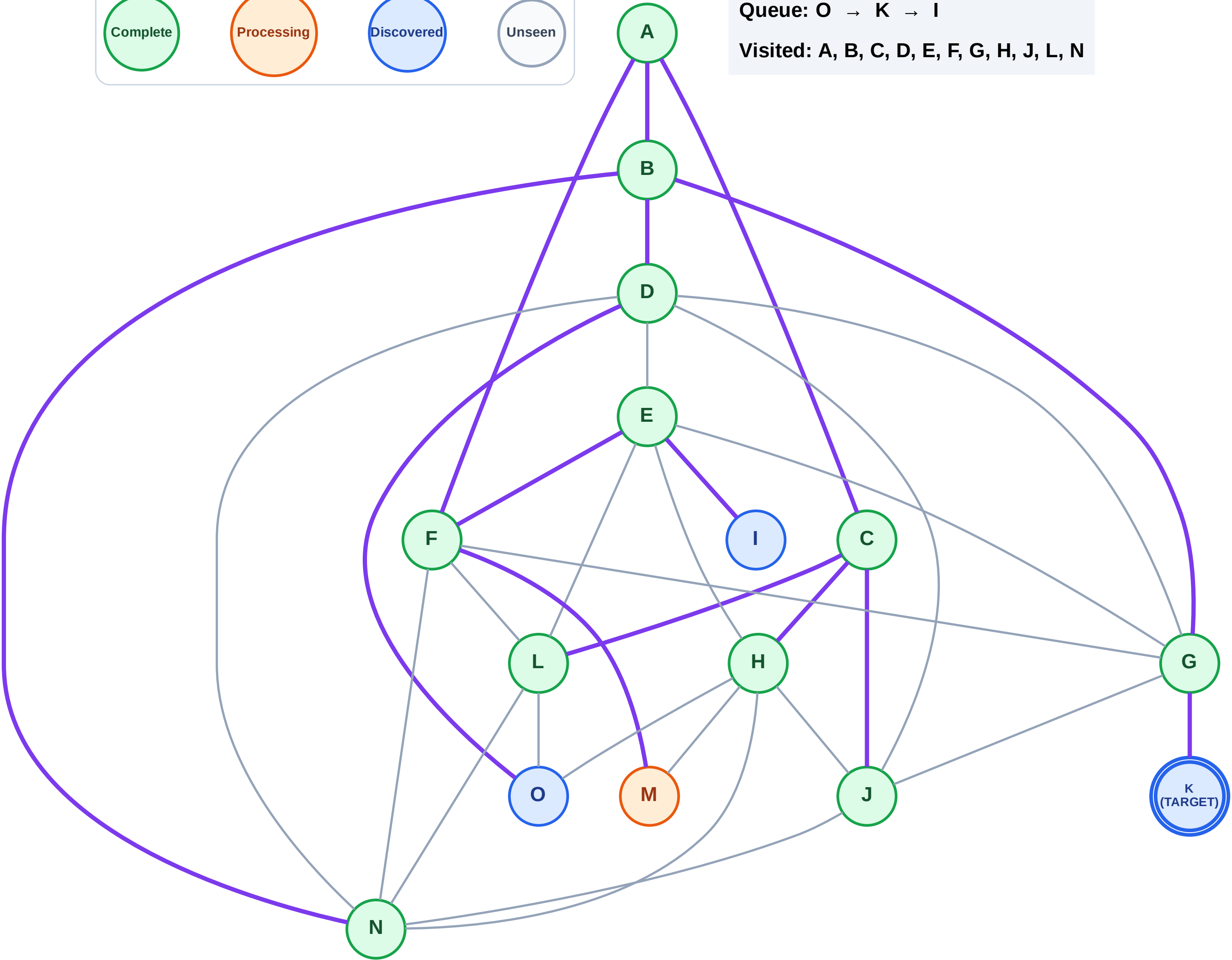
Processing

Discovered

Unseen

Queue: O → K → I

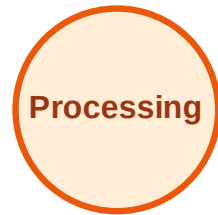
Visited: A, B, C, D, E, F, G, H, J, L, N



BFS Traversal

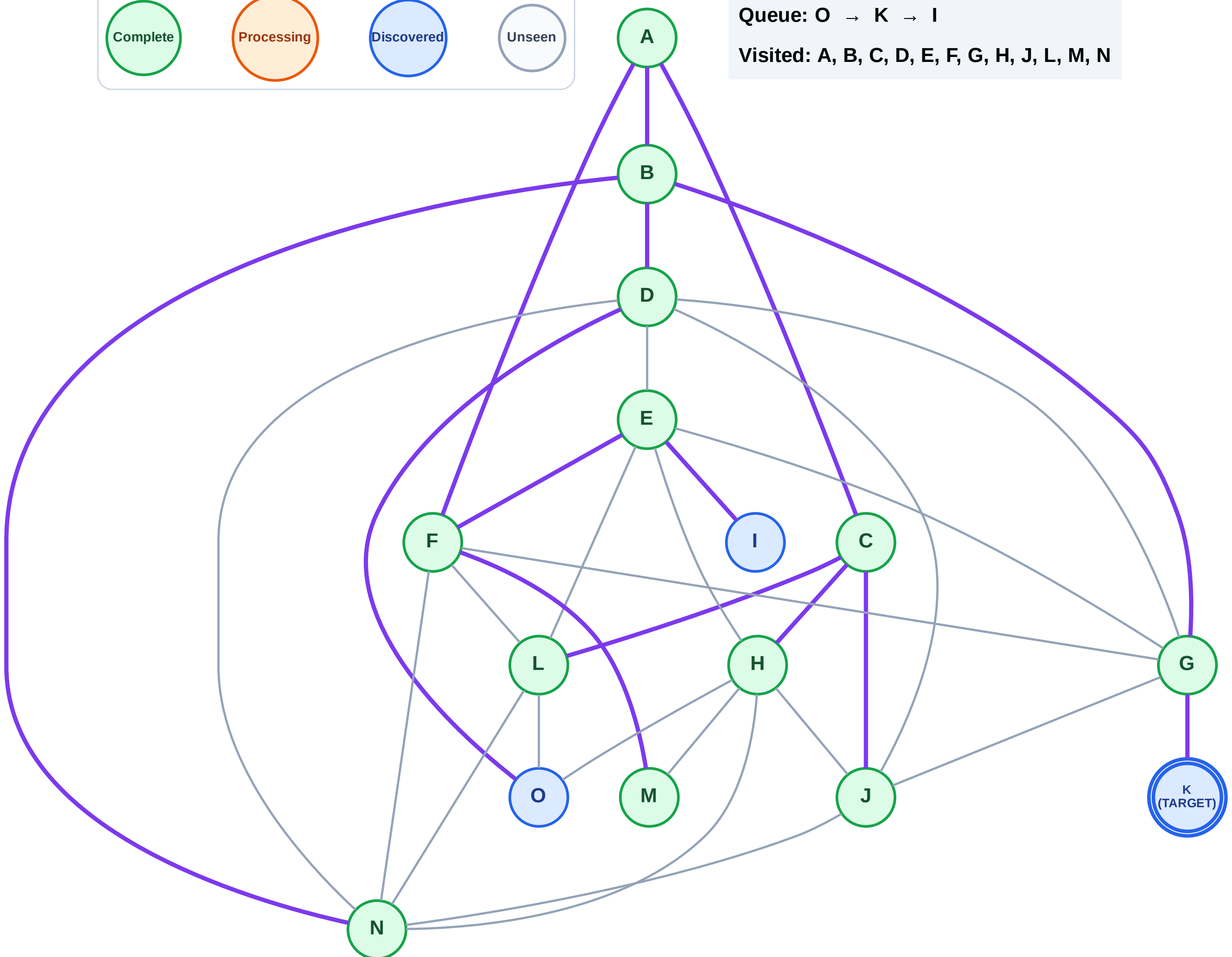
Step 25: No new neighbors from M

Legend



Queue: O → K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, N



BFS Traversal

Step 26: Process node O

Legend

Complete

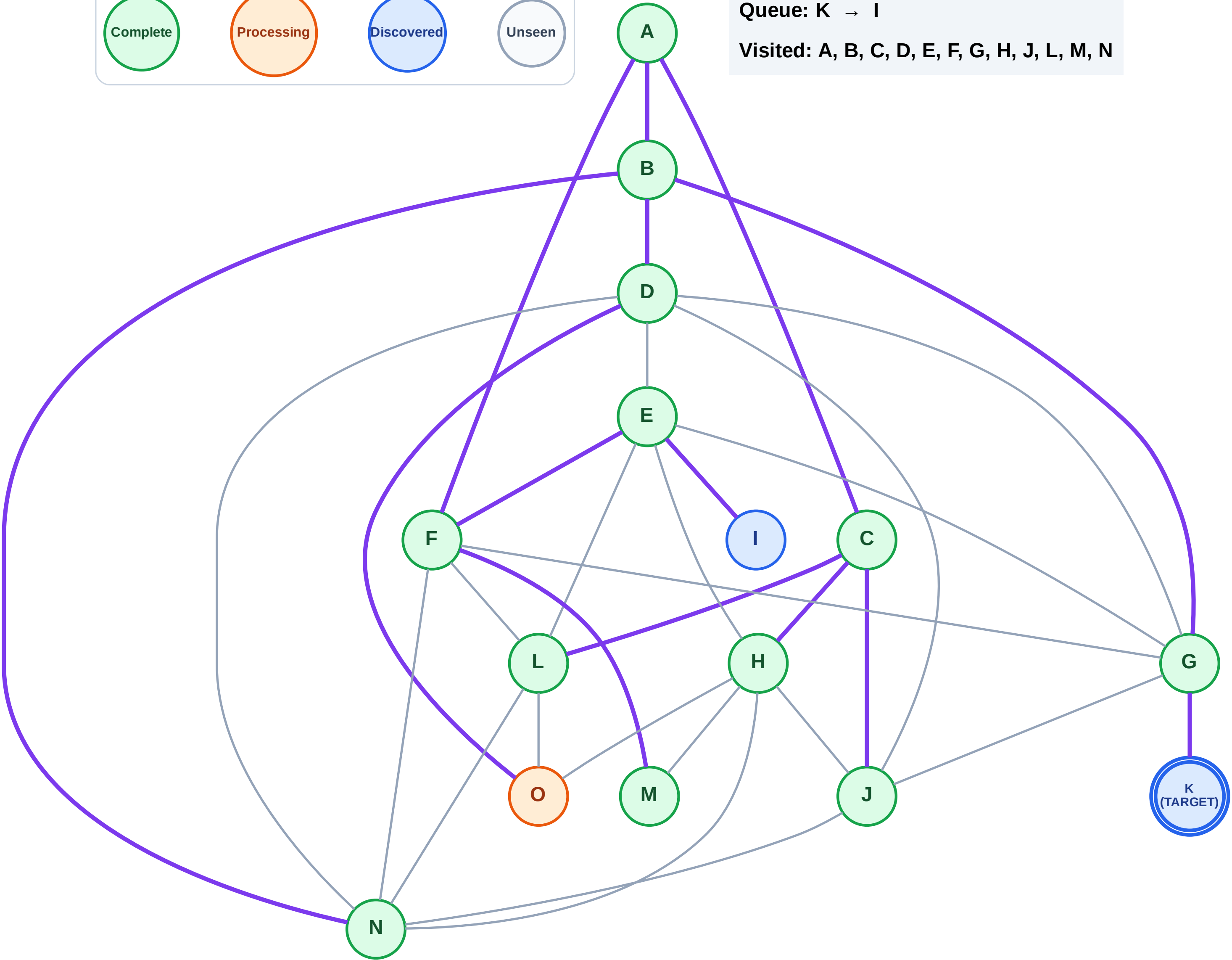
Processing

Discovered

Unseen

Queue: K → I

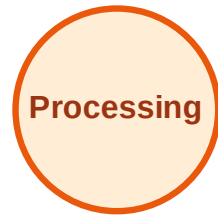
Visited: A, B, C, D, E, F, G, H, J, L, M, N



BFS Traversal

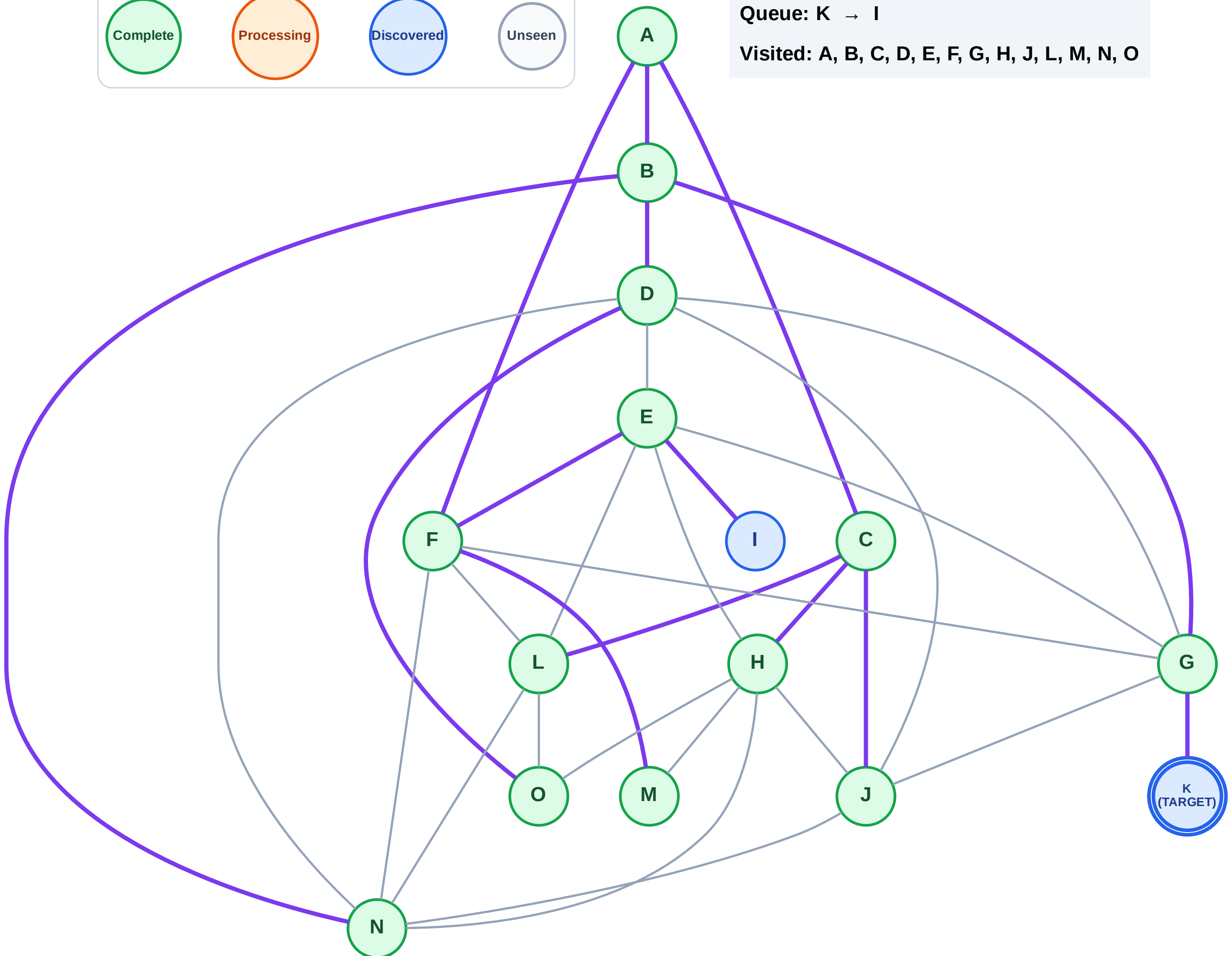
Step 27: No new neighbors from O

Legend



Queue: K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

Step 28: Process node K

Legend

Complete

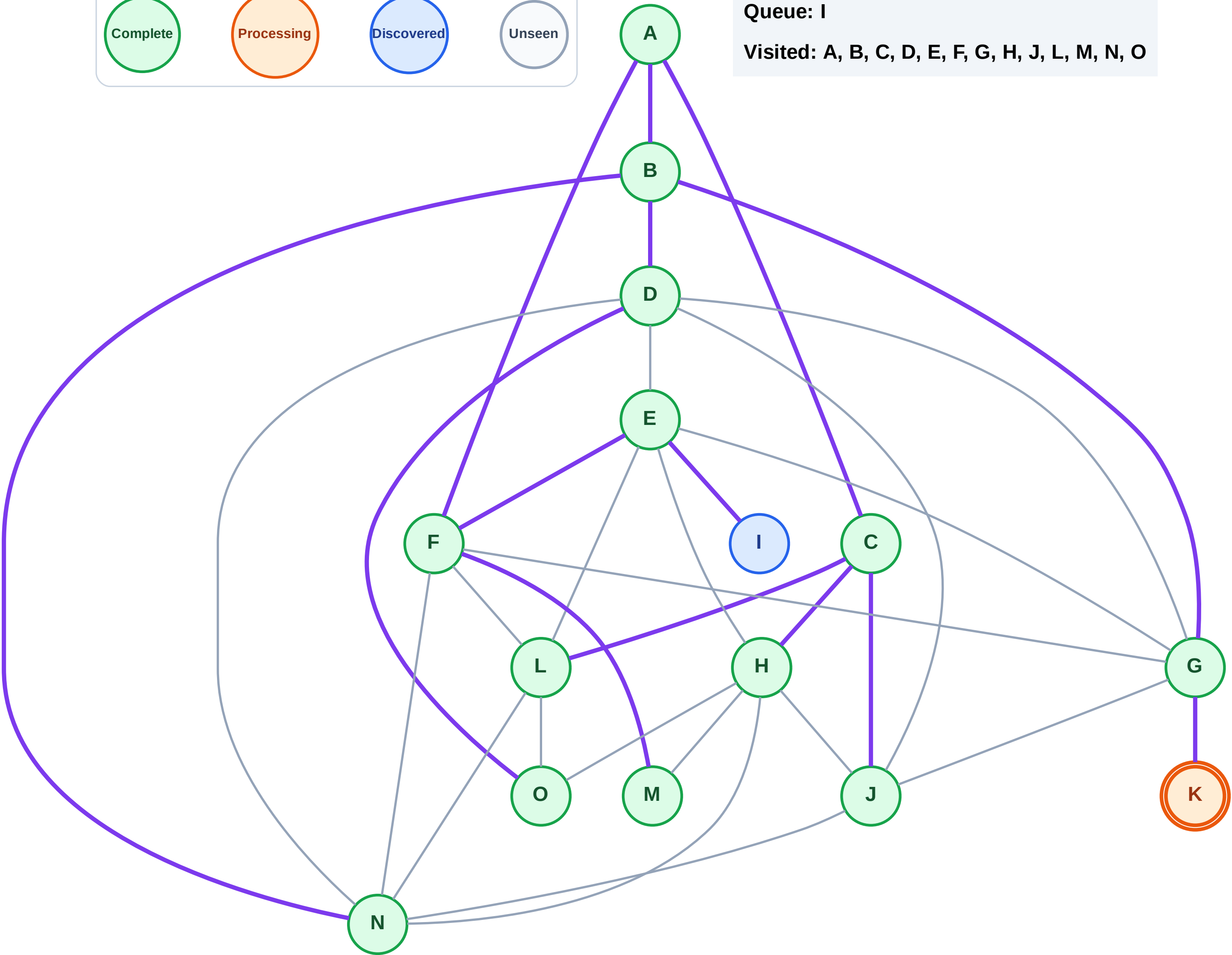
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → B → G → K

Legend

Complete

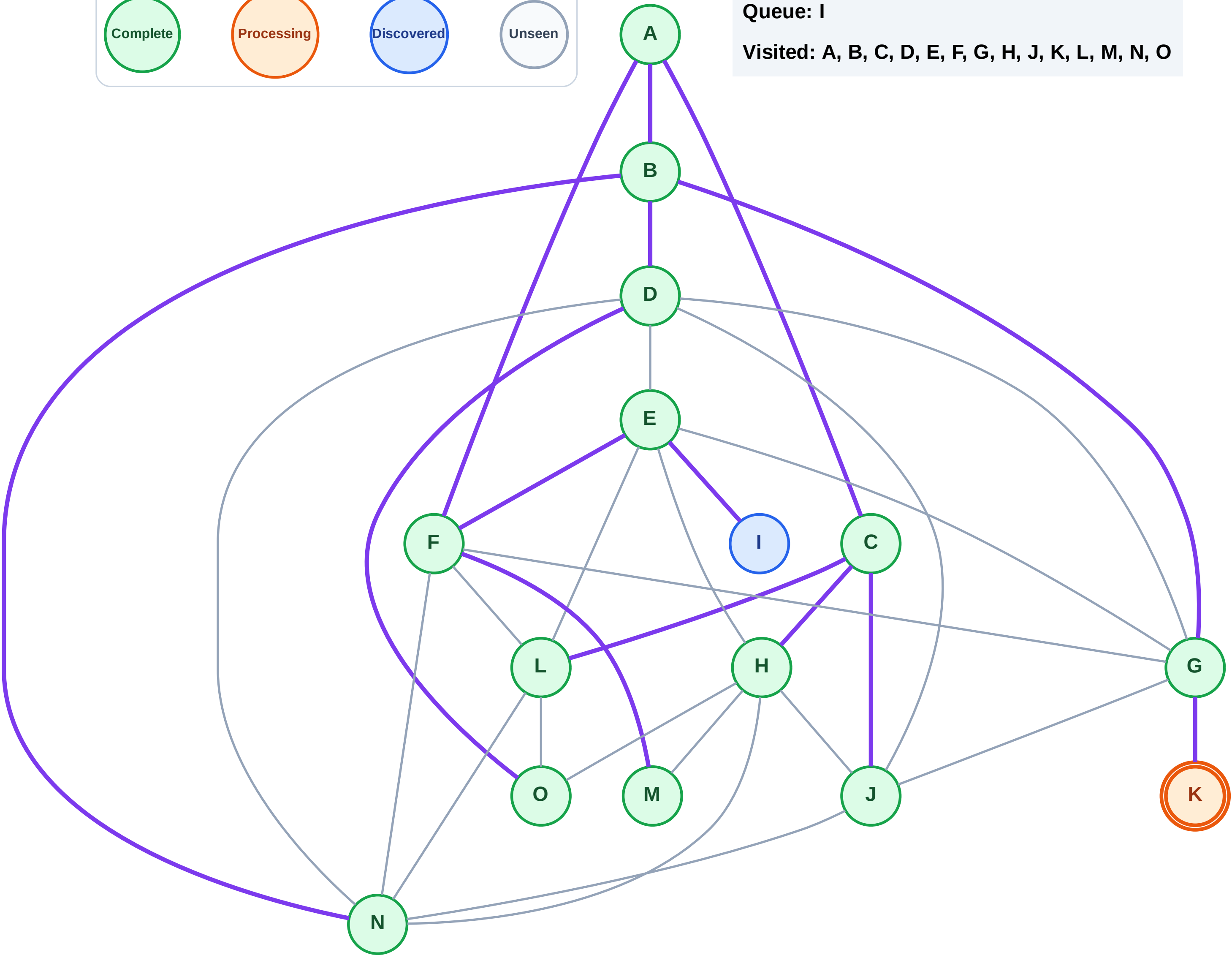
Processing

Discovered

Unseen

Queue: I

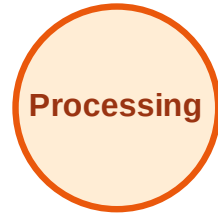
Visited: A, B, C, D, E, F, G, H, J, K, L, M, N, O



DFS Traversal

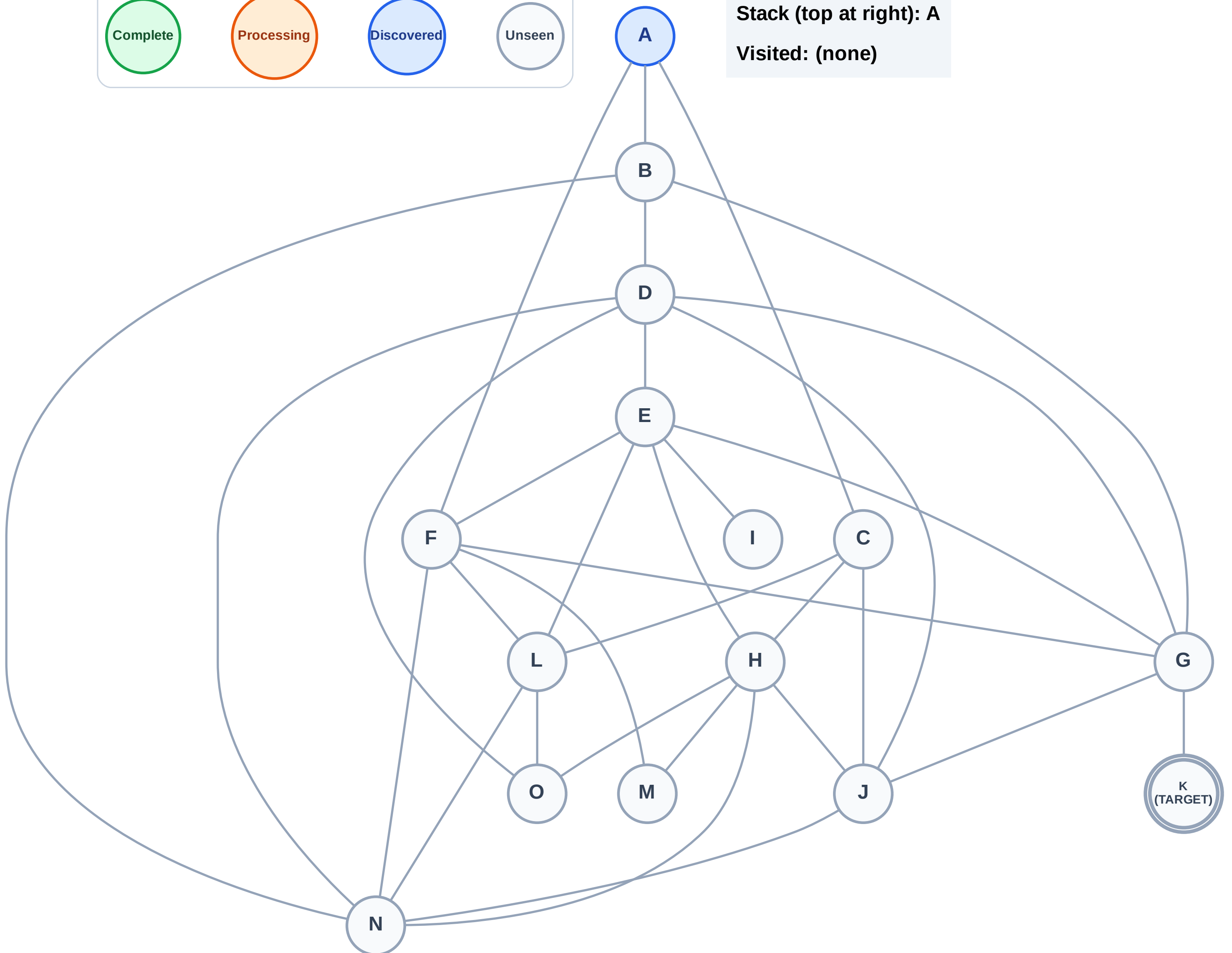
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

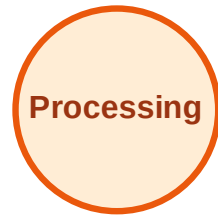
Visited: (none)



DFS Traversal

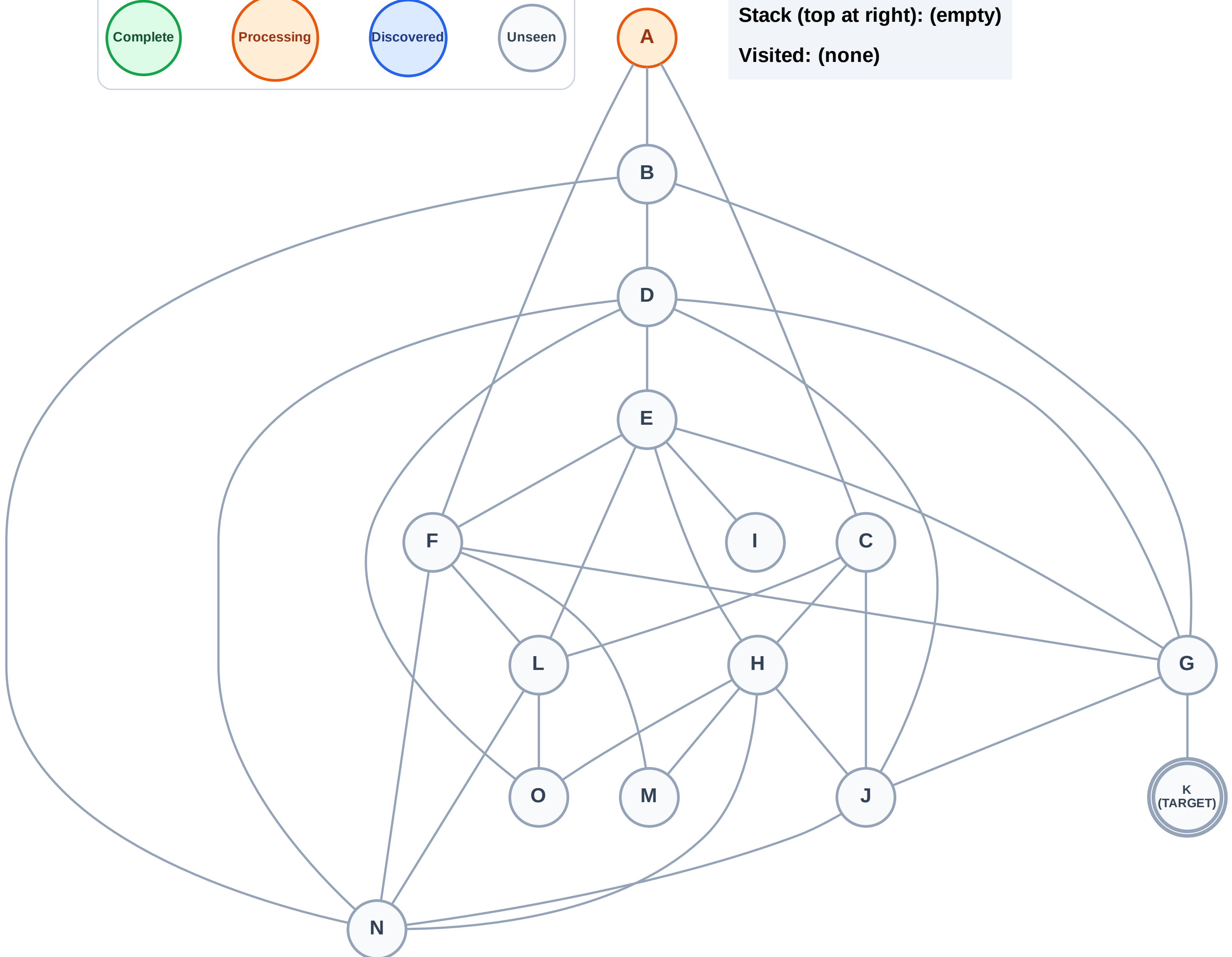
Step 2: Process node A

Legend



Stack (top at right): (empty)

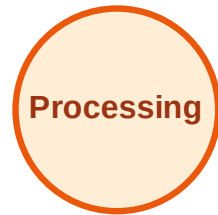
Visited: (none)



DFS Traversal

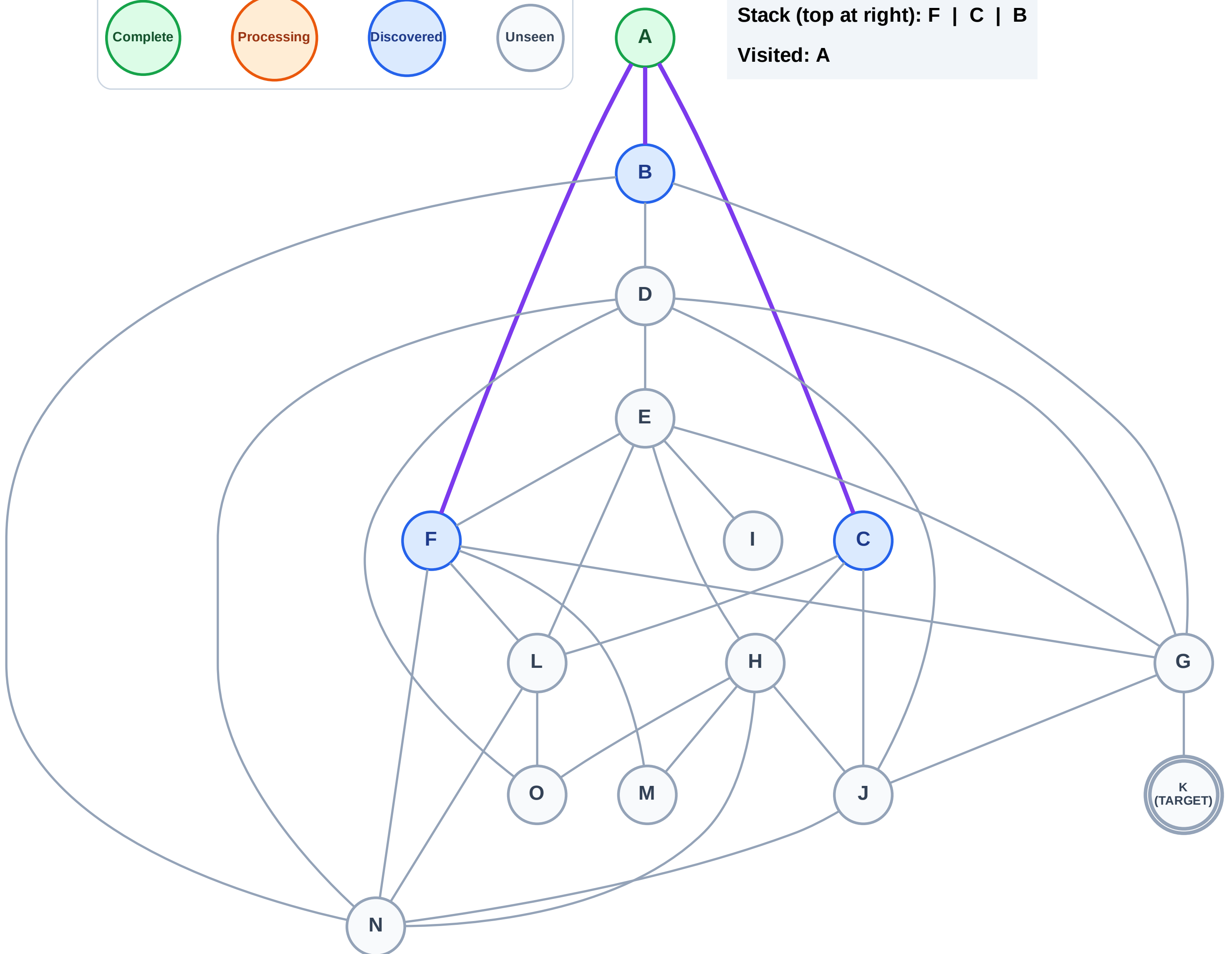
Step 3: Discover F, C, B from A

Legend



Stack (top at right): F | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

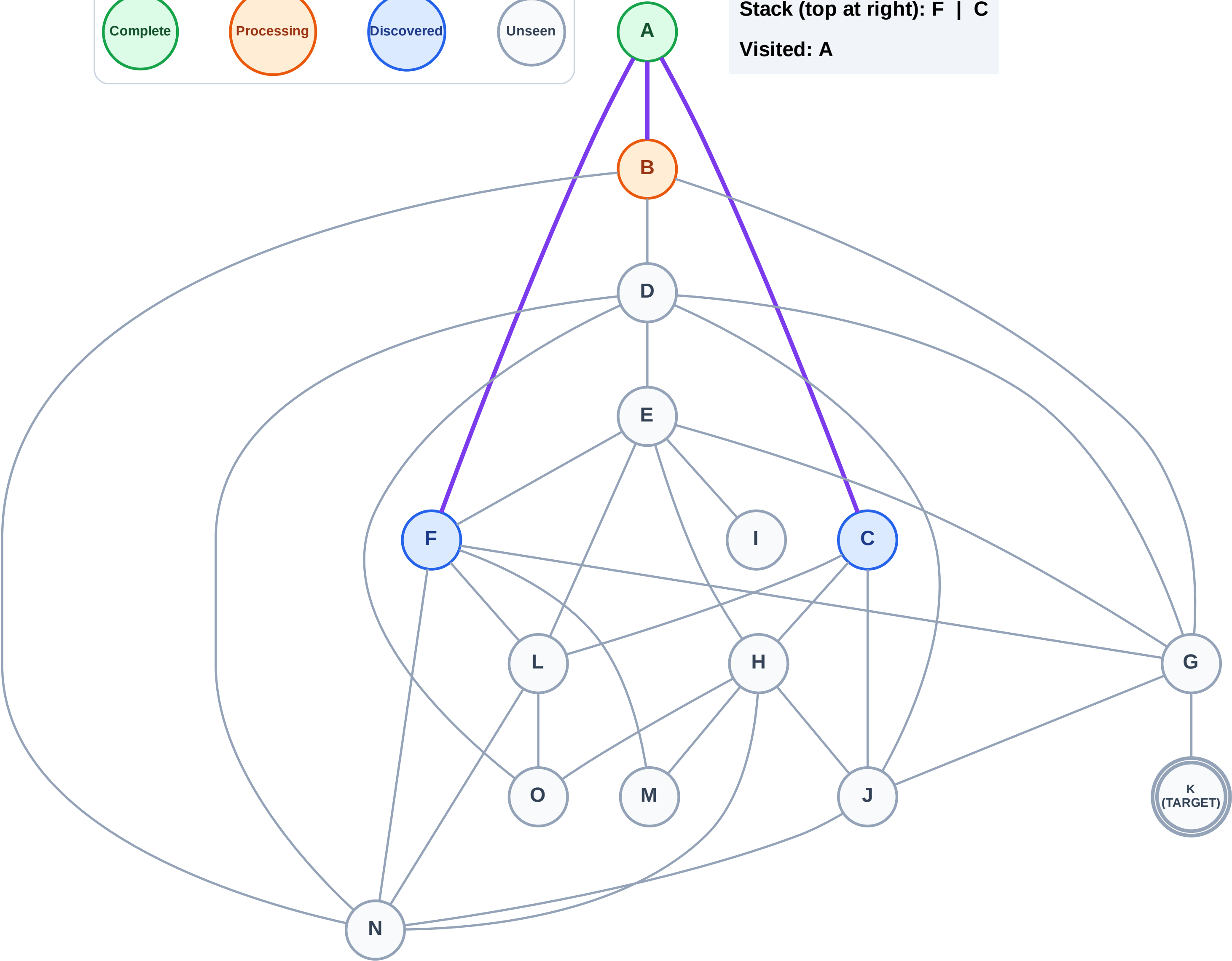
Processing

Discovered

Unseen

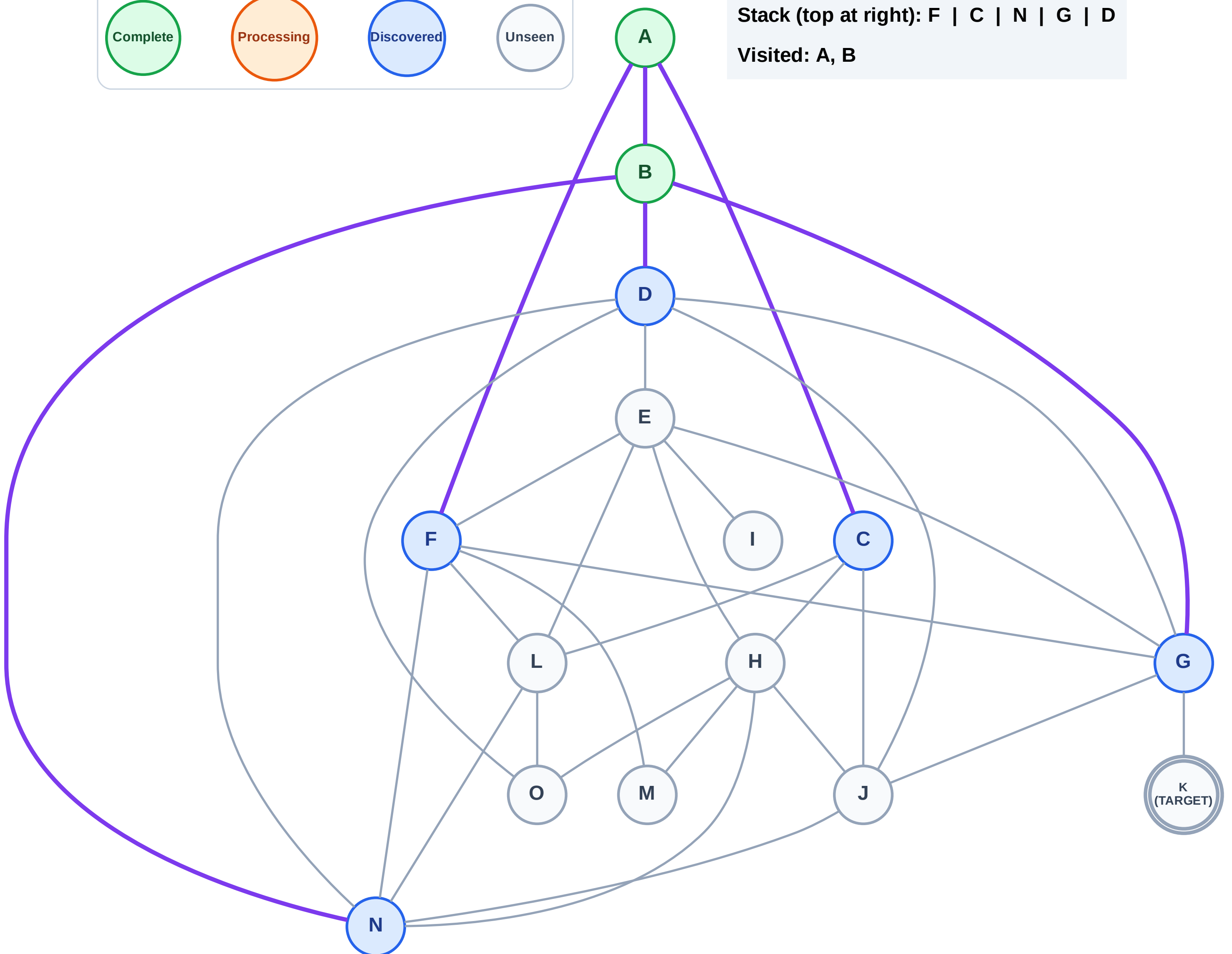
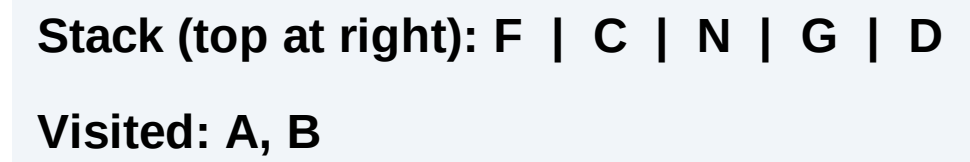
Stack (top at right): F | C

Visited: A



Step 5: Discover N, G, D from B

Step 5: Discover N, G, D from B



DFS Traversal

Step 6: Process node D

Legend

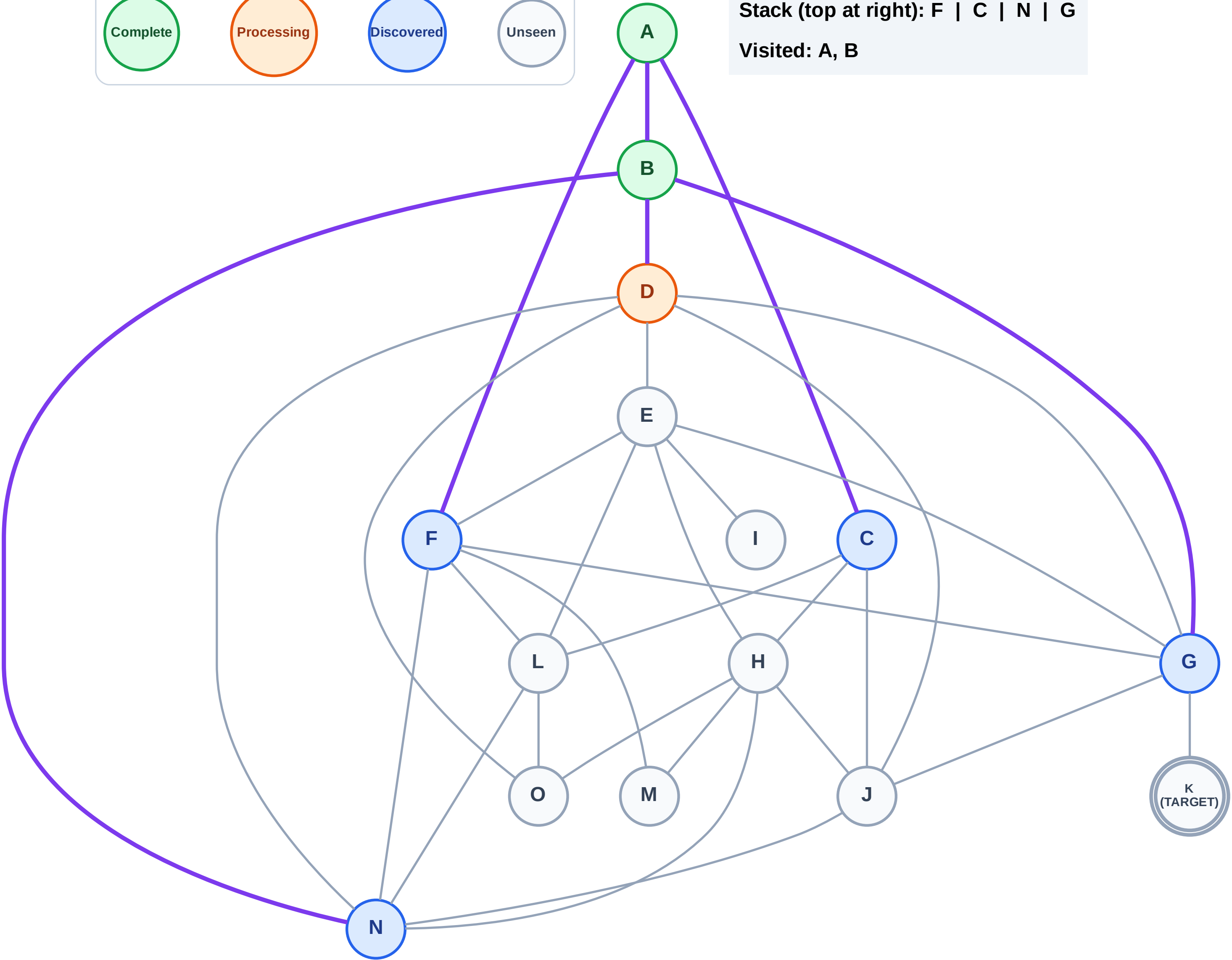
Complete

Processing

Discovered

Unseen

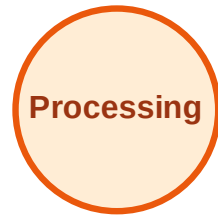
Stack (top at right): F | C | N | G
Visited: A, B



DFS Traversal

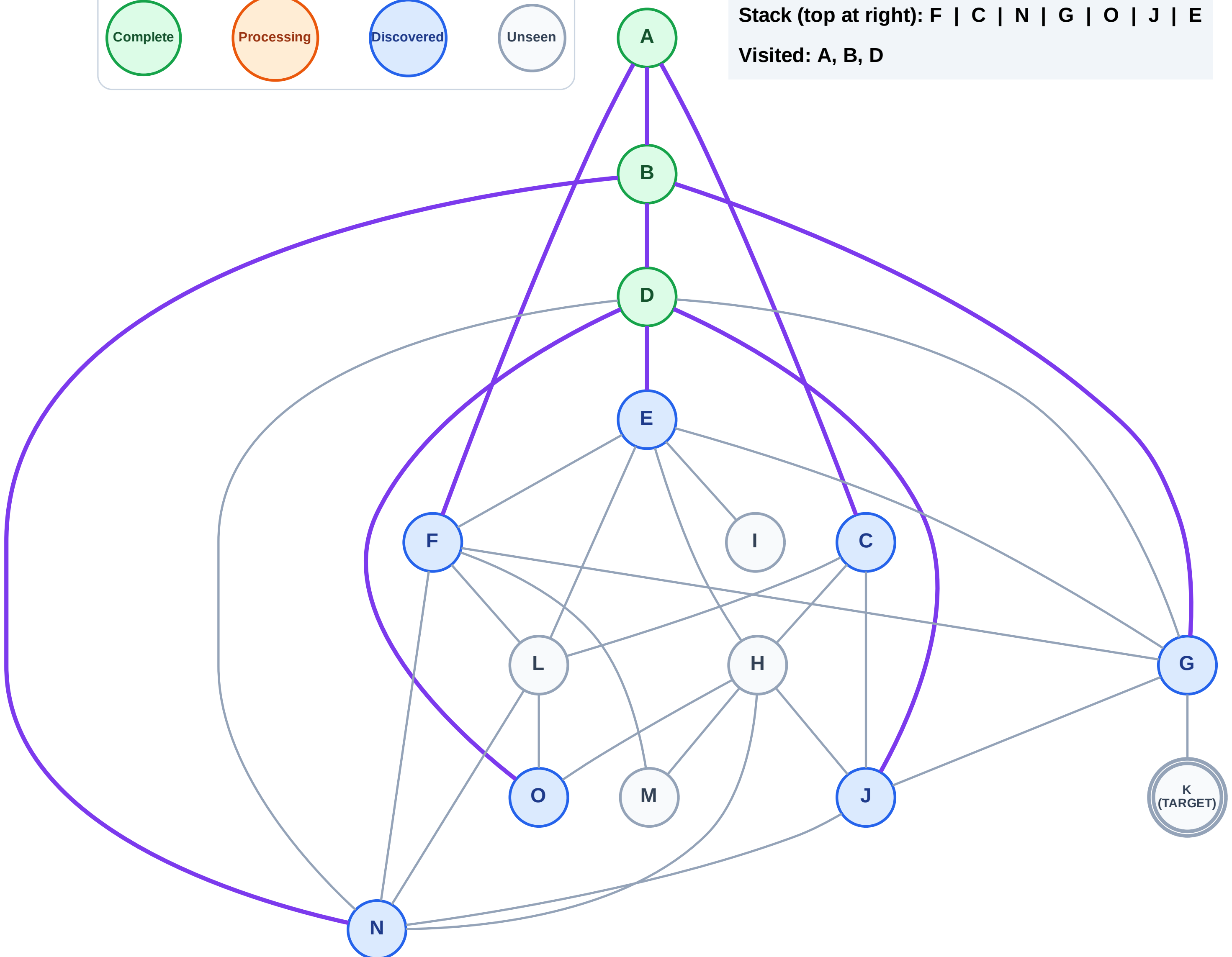
Step 7: Discover O, J, E from D

Legend



Stack (top at right): F | C | N | G | O | J | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

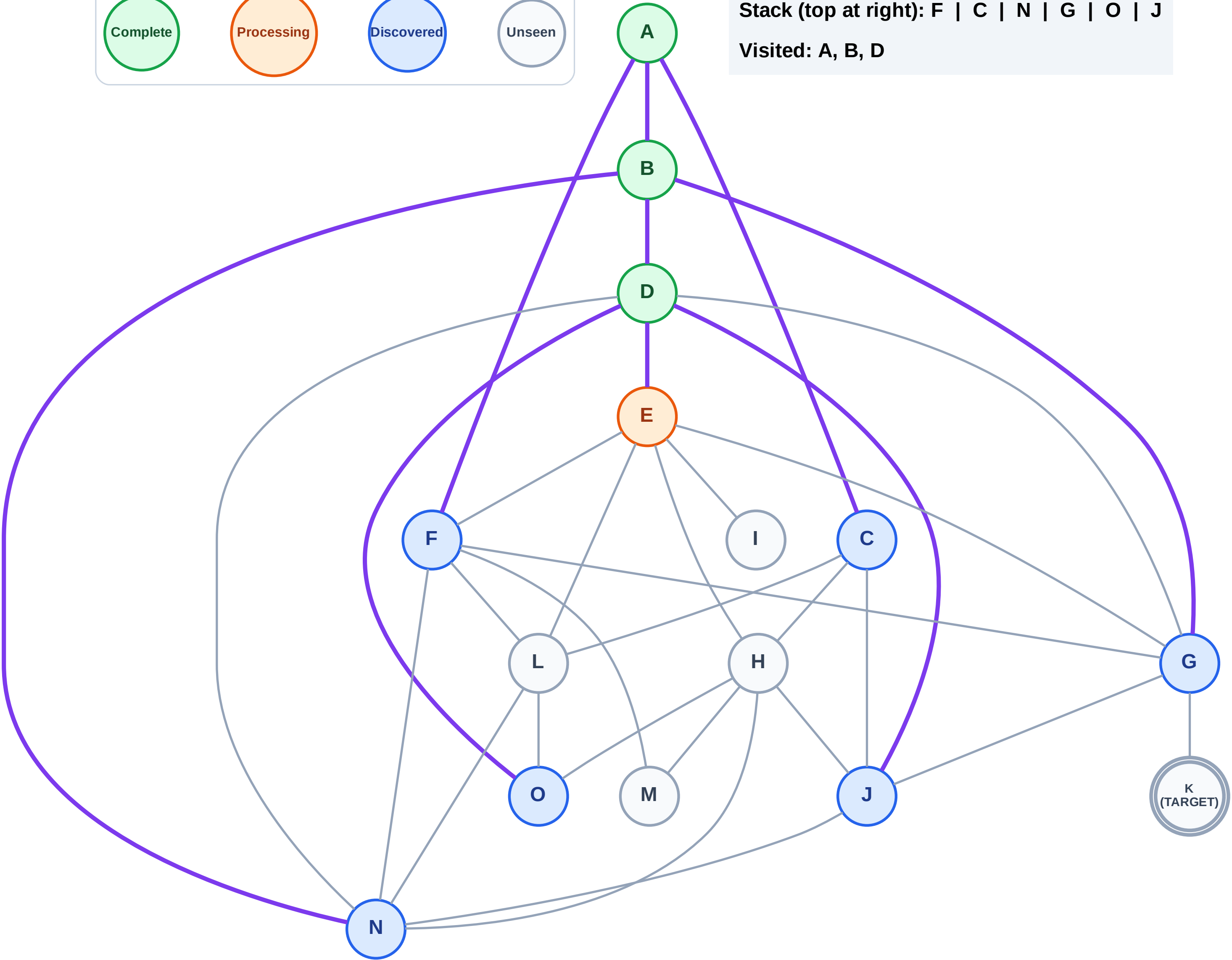
Processing

Discovered

Unseen

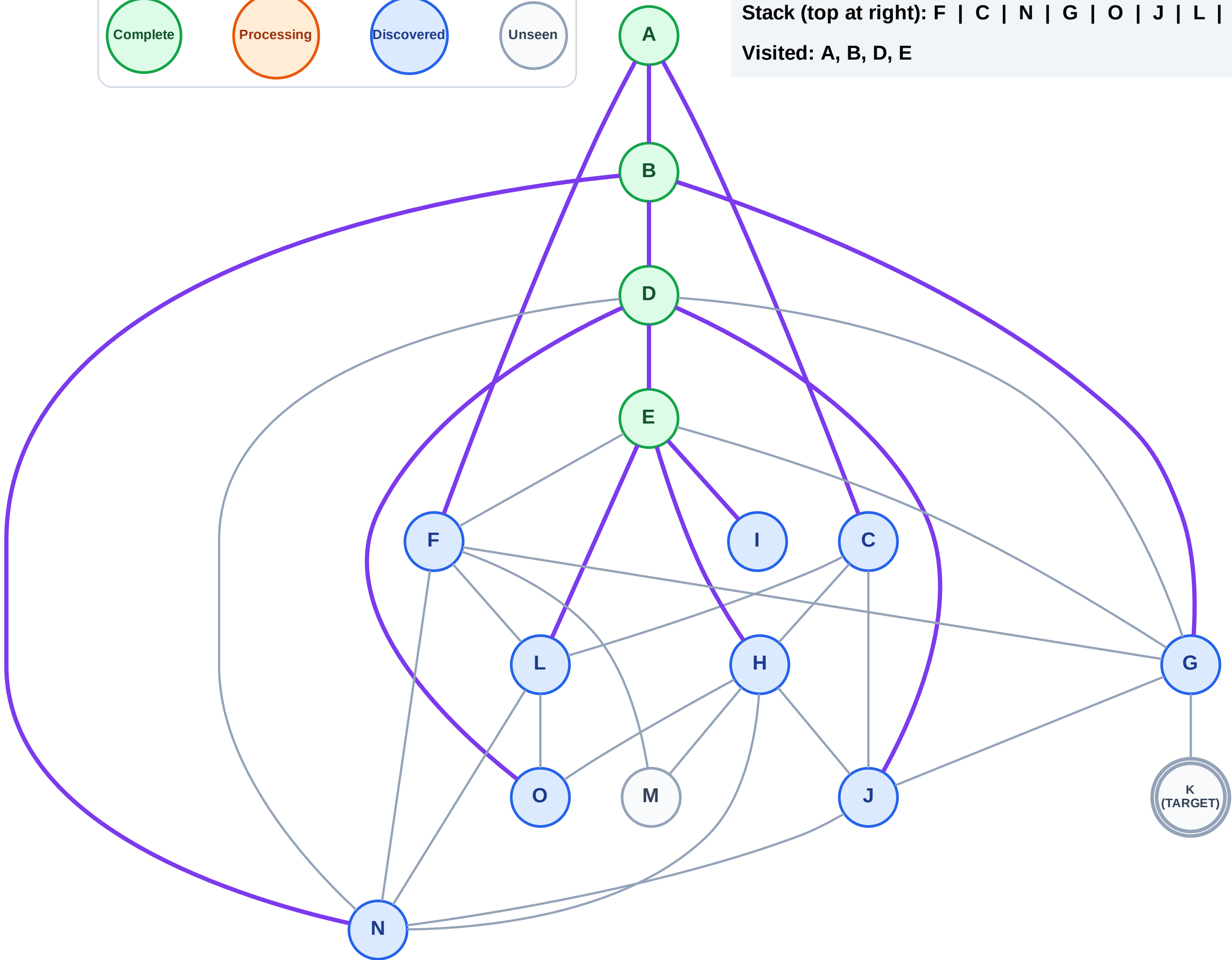
Stack (top at right): F | C | N | G | O | J

Visited: A, B, D



Step 9: Discover L, I, H from E

Step 9: Discover L, I, H from E



DFS Traversal

Step 10: Process node H

Legend

Complete

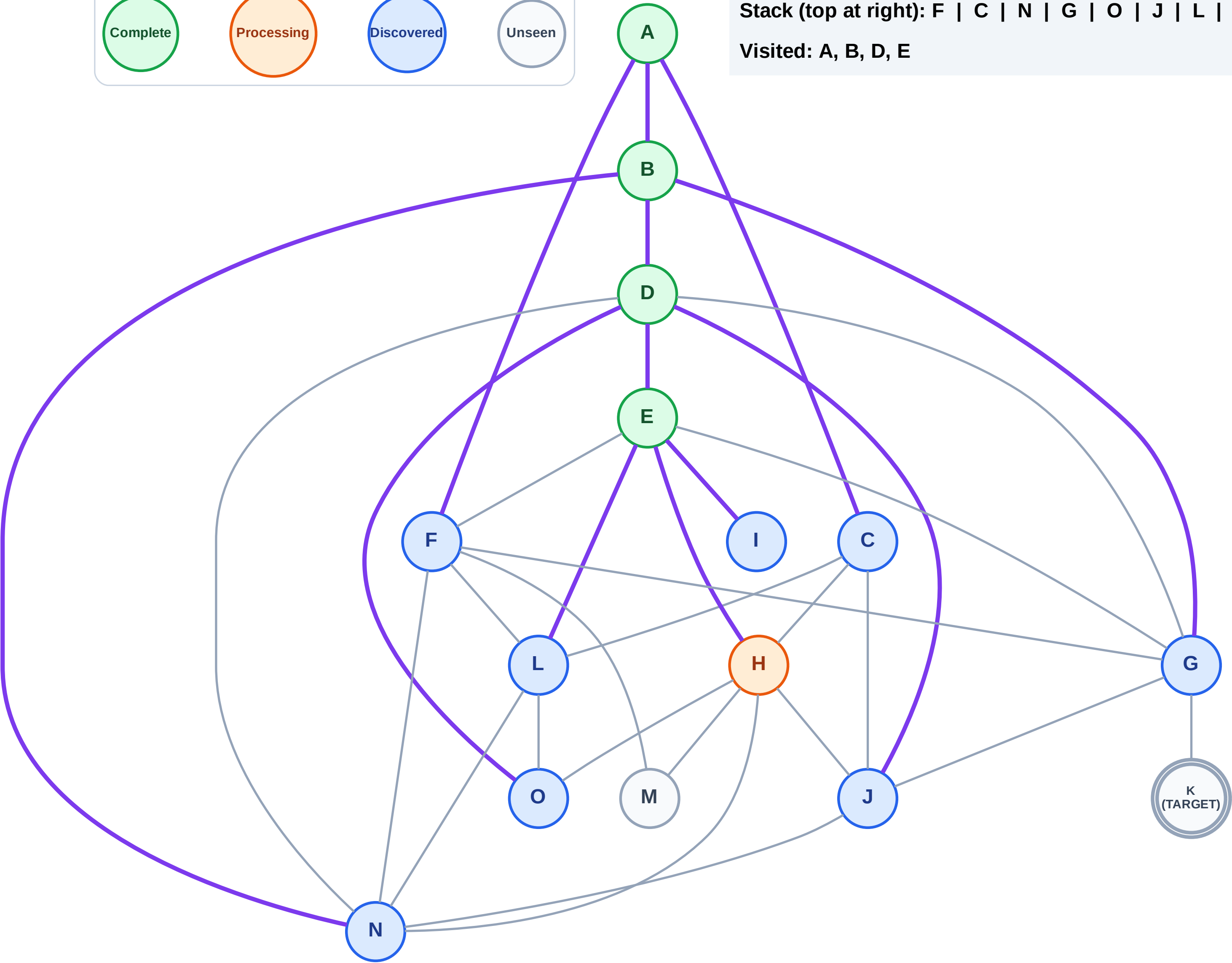
Processing

Discovered

Unseen

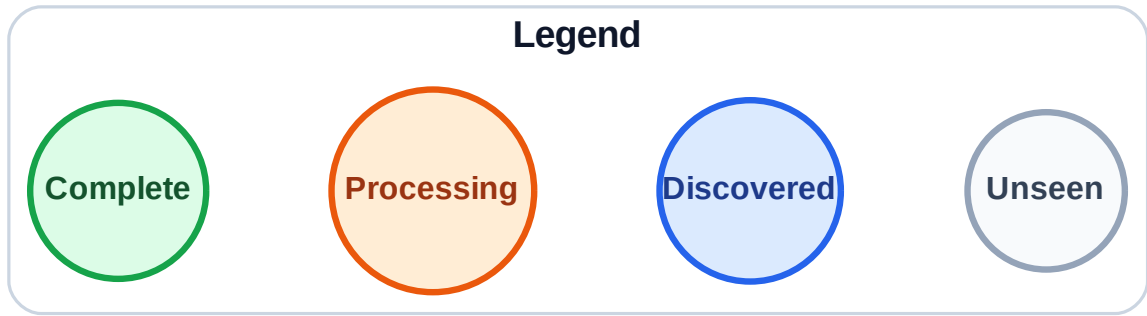
Stack (top at right): F | C | N | G | O | J | L | I

Visited: A, B, D, E



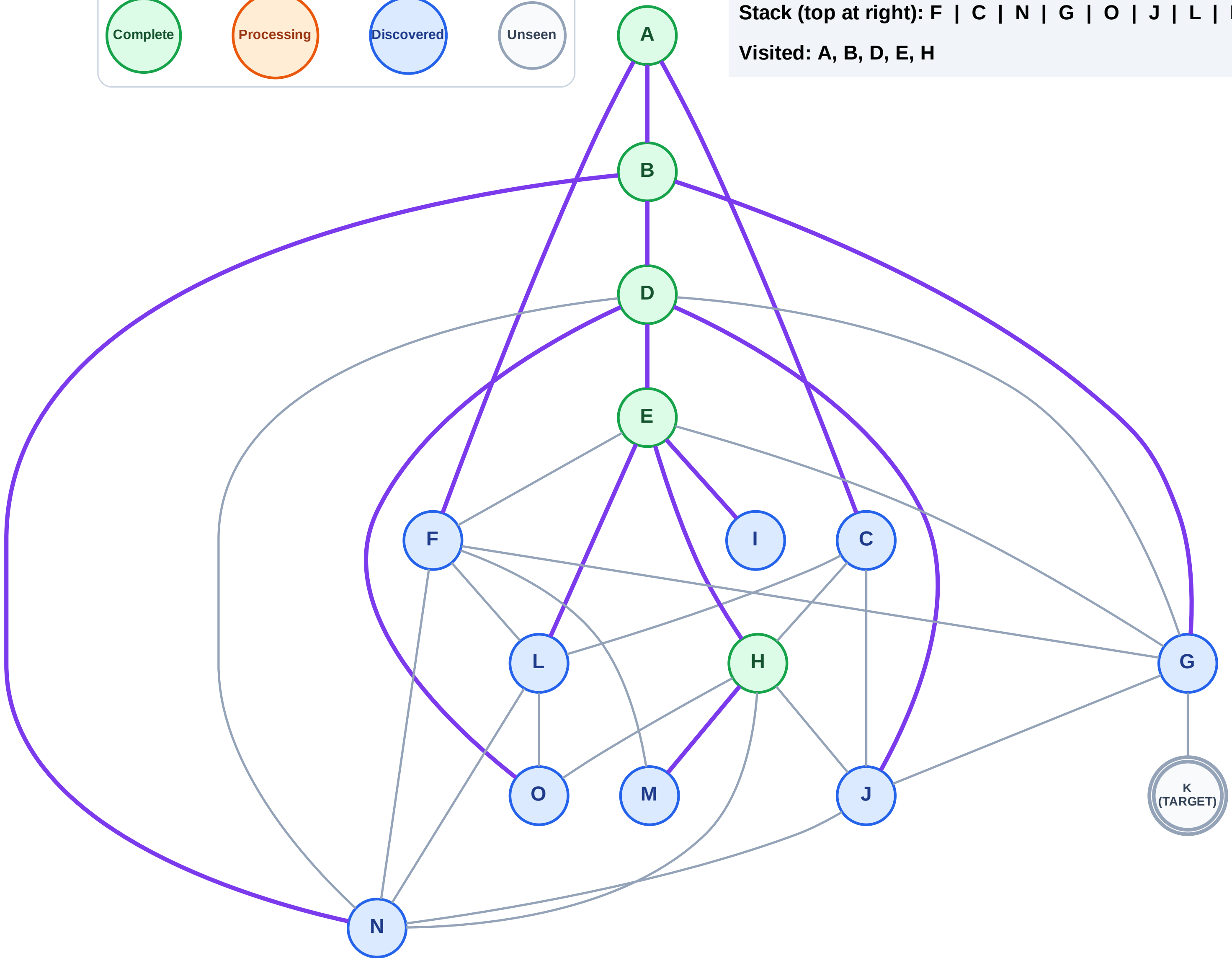
DFS Traversal

Step 11: Discover M from H



Stack (top at right): F | C | N | G | O | J | L | I | M

Visited: A, B, D, E, H



DFS Traversal

Step 12: Process node M

Legend

Complete

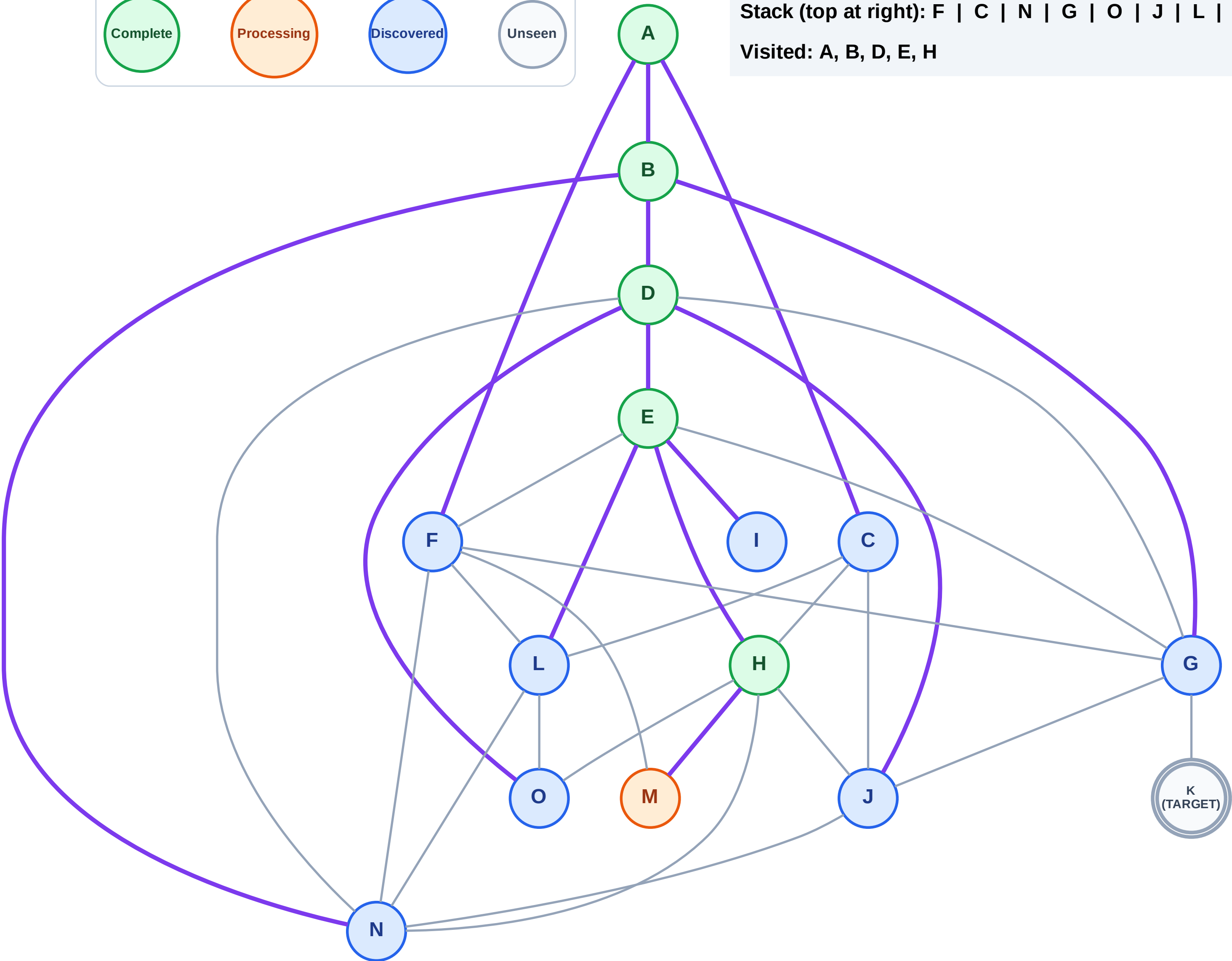
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J | L | I

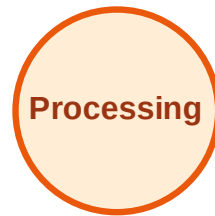
Visited: A, B, D, E, H



DFS Traversal

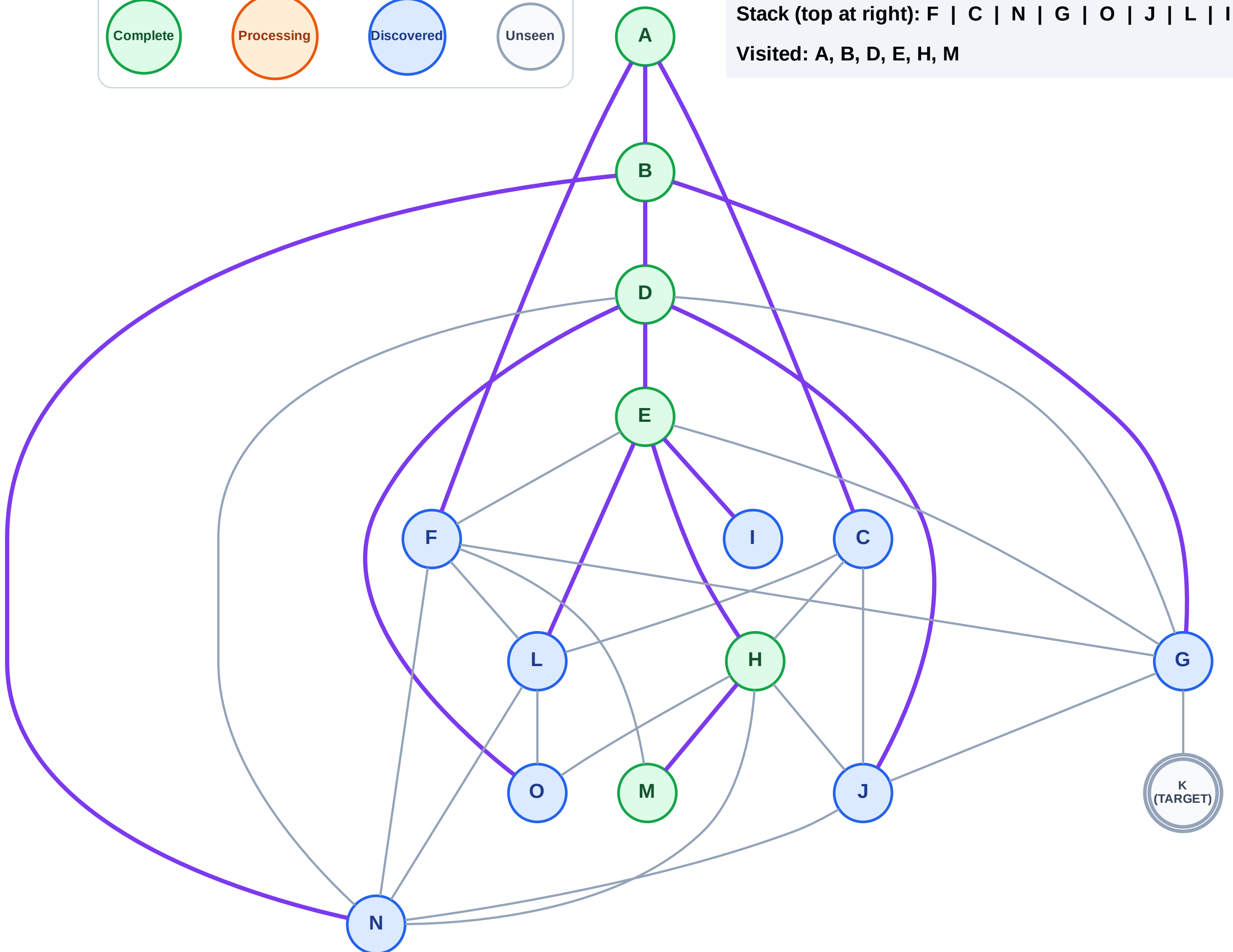
Step 13: No new neighbors from M

Legend



Stack (top at right): F | C | N | G | O | J | L | I

Visited: A, B, D, E, H, M



Step 14: Process node I

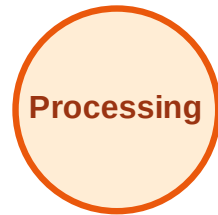
Step 14: Process node I



DFS Traversal

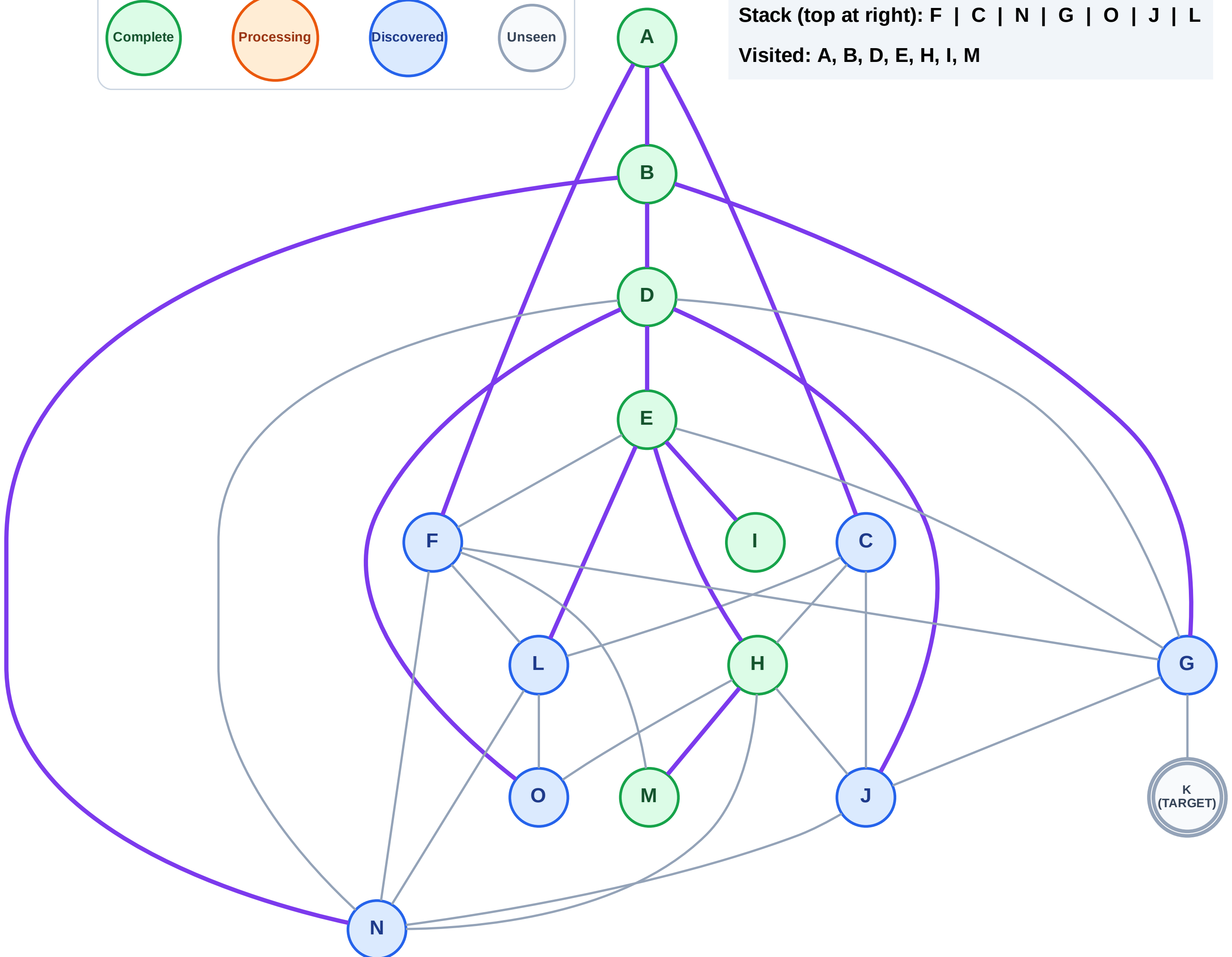
Step 15: No new neighbors from I

Legend



Stack (top at right): F | C | N | G | O | J | L

Visited: A, B, D, E, H, I, M



DFS Traversal

Step 16: Process node L

Legend

Complete

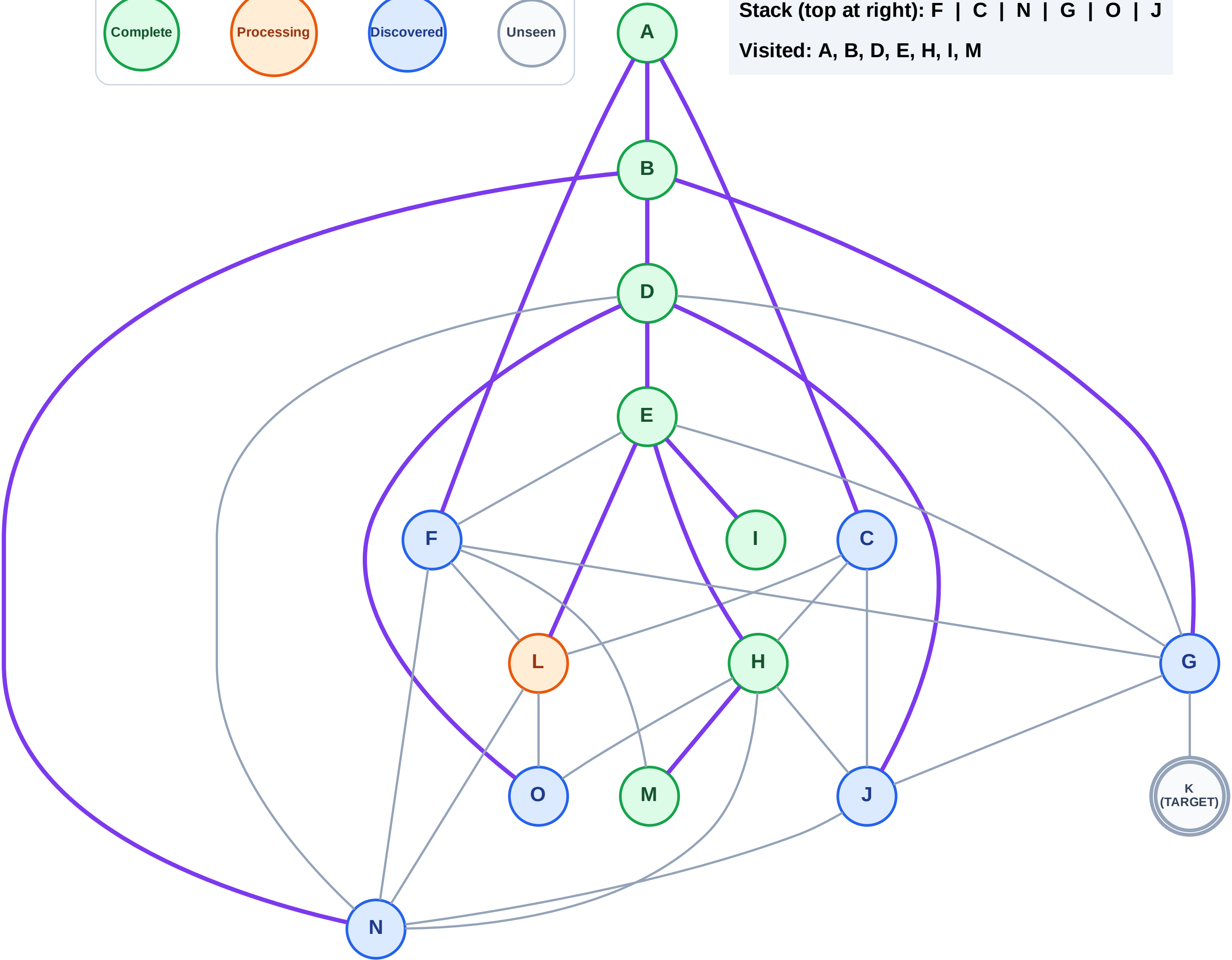
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J

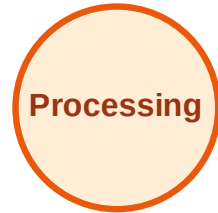
Visited: A, B, D, E, H, I, M



DFS Traversal

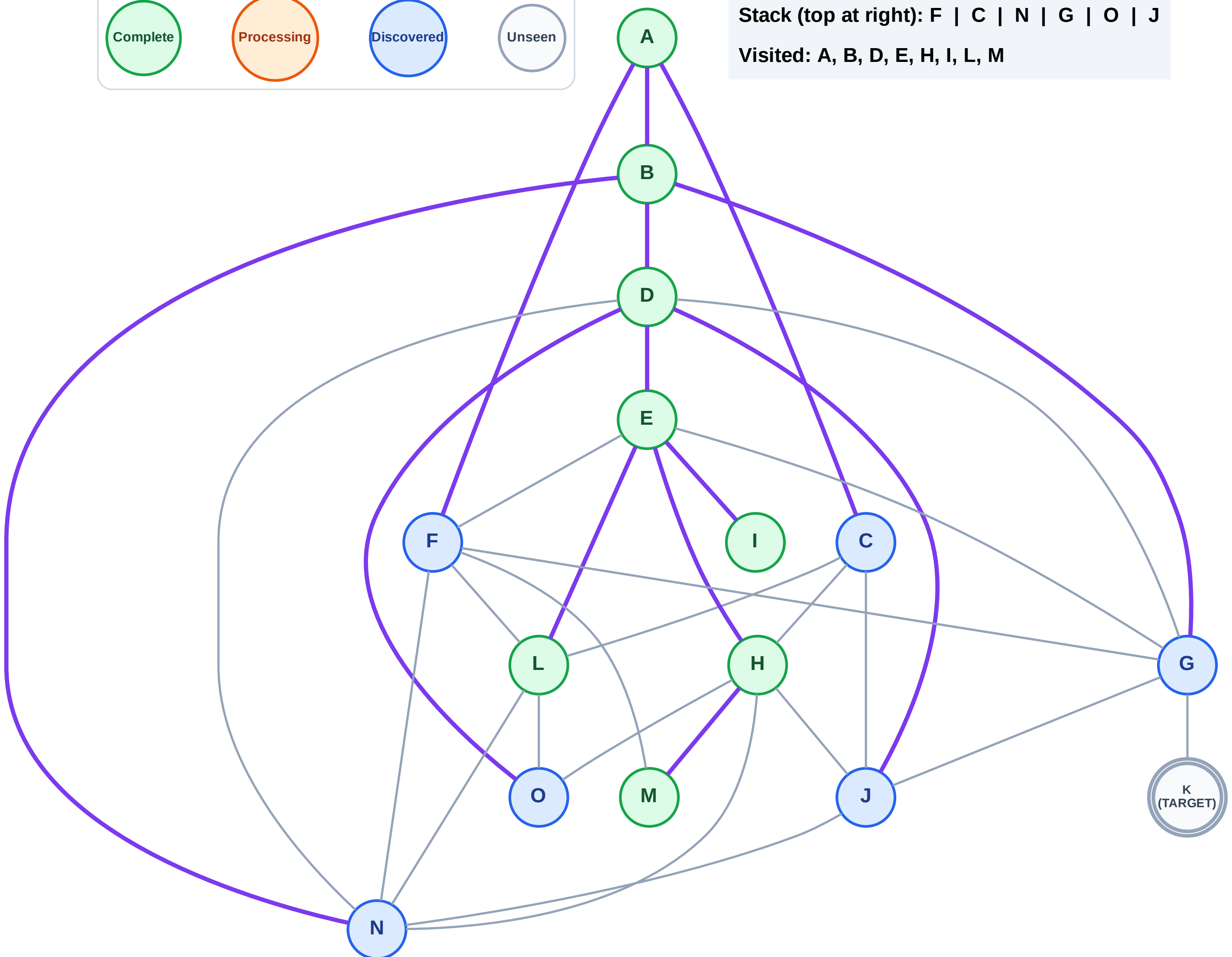
Step 17: No new neighbors from L

Legend



Stack (top at right): F | C | N | G | O | J

Visited: A, B, D, E, H, I, L, M



DFS Traversal

Step 18: Process node J

Legend

Complete

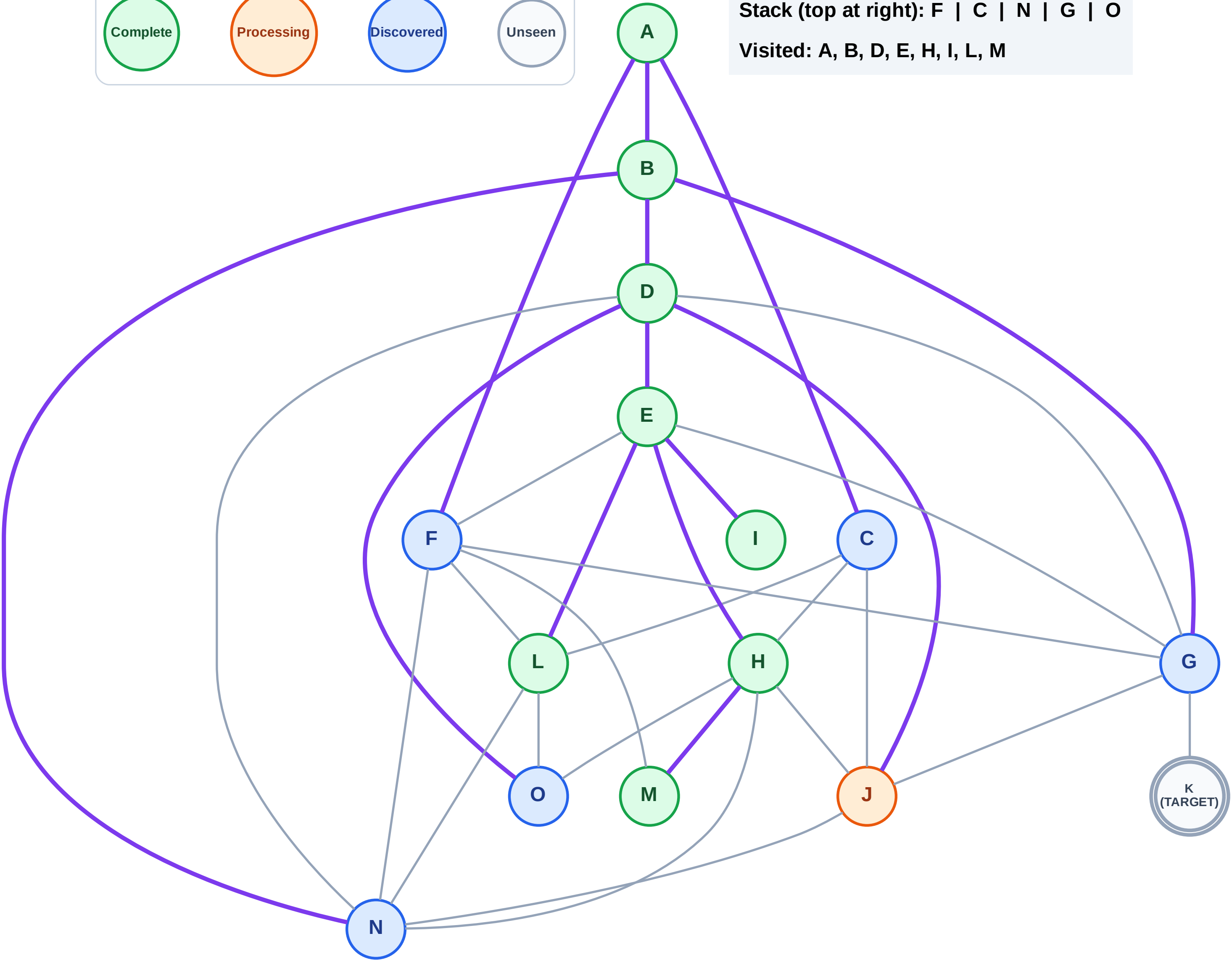
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O

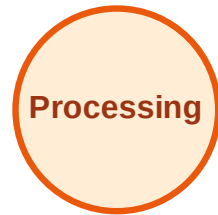
Visited: A, B, D, E, H, I, L, M



DFS Traversal

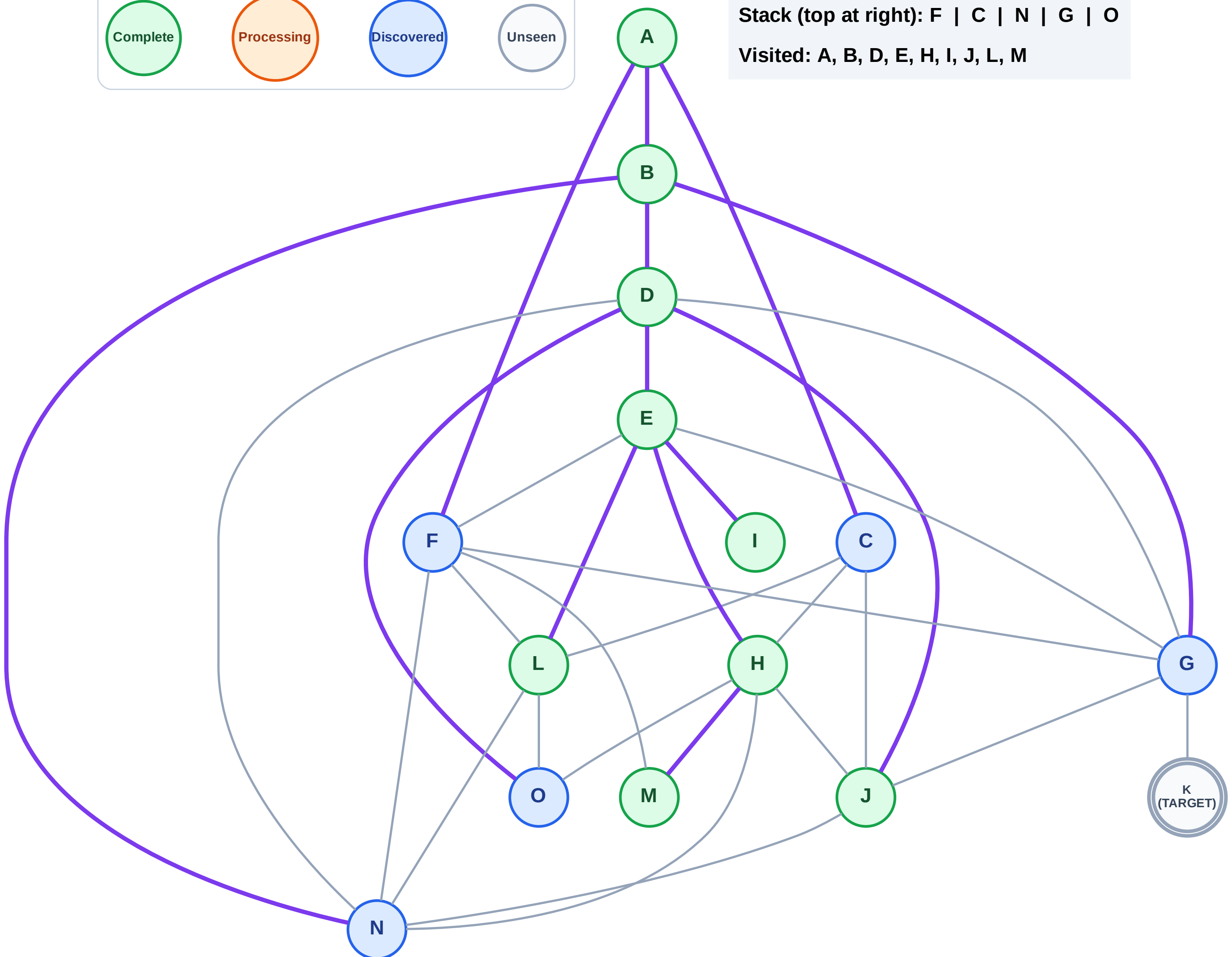
Step 19: No new neighbors from J

Legend



Stack (top at right): F | C | N | G | O

Visited: A, B, D, E, H, I, J, L, M



DFS Traversal

Step 20: Process node O

Legend

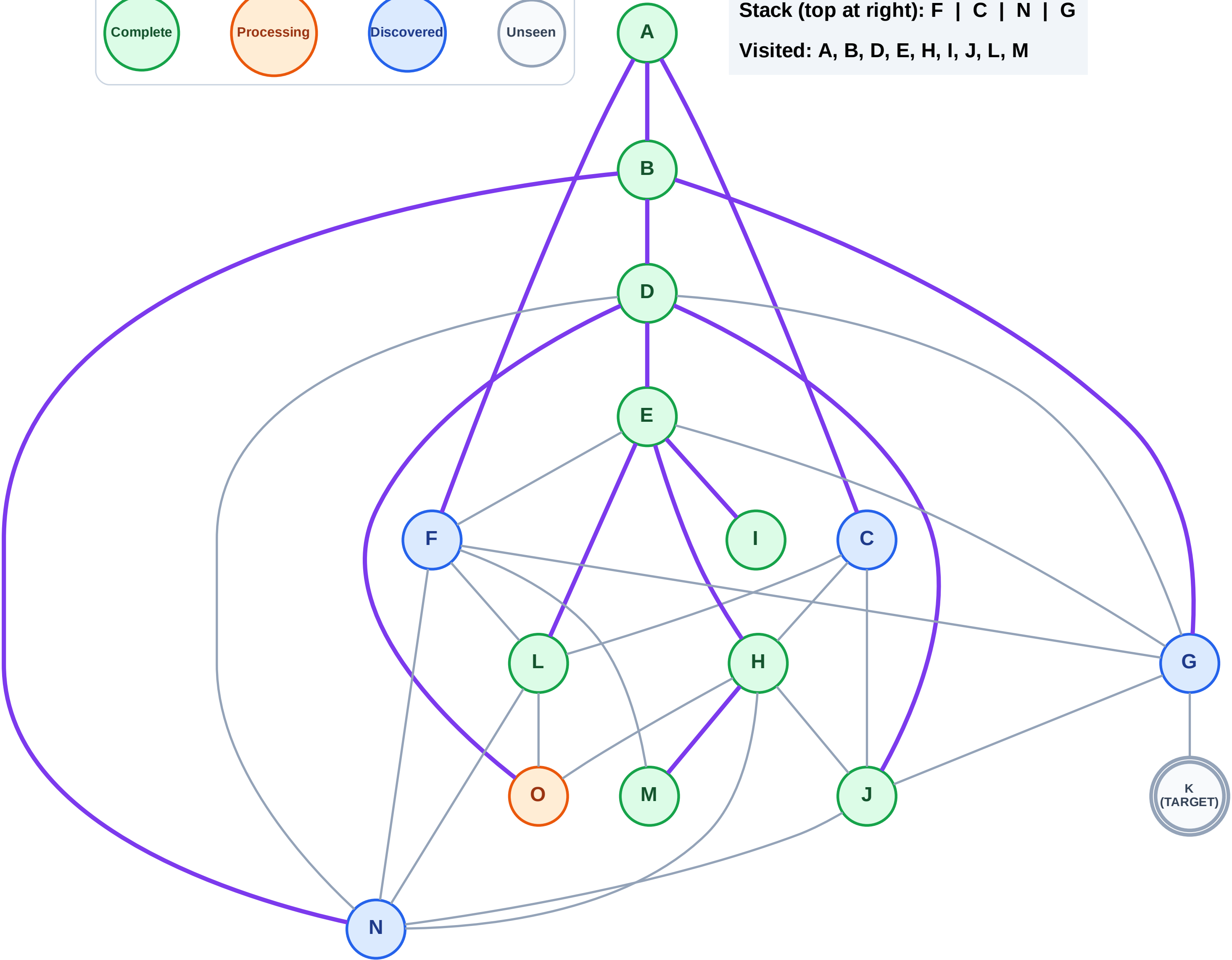
Complete

Processing

Discovered

Unseen

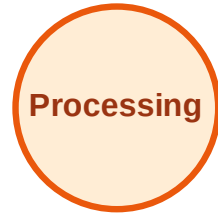
Stack (top at right): F | C | N | G
Visited: A, B, D, E, H, I, J, L, M



DFS Traversal

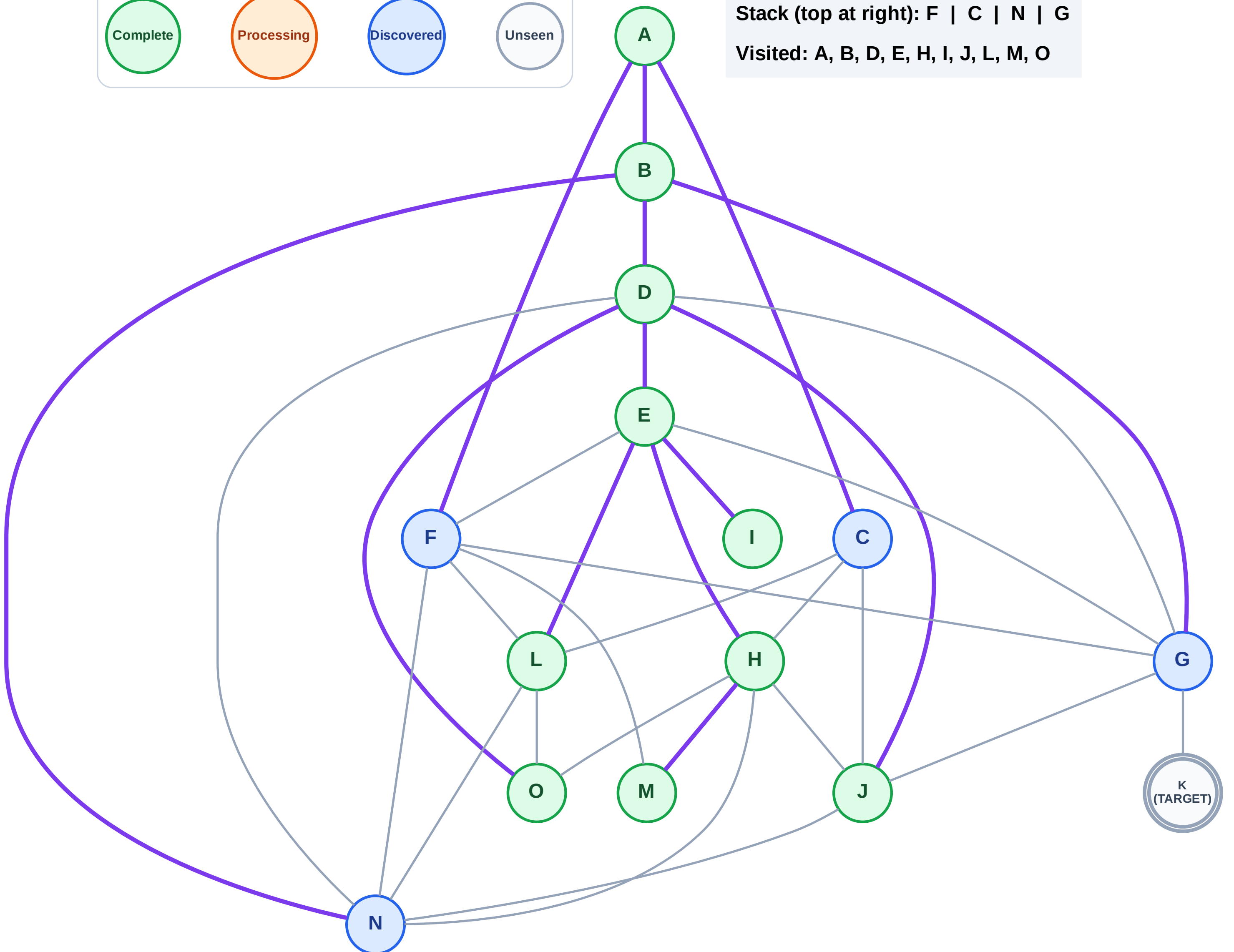
Step 21: No new neighbors from O

Legend



Stack (top at right): F | C | N | G

Visited: A, B, D, E, H, I, J, L, M, O



DFS Traversal

Step 22: Process node G

Legend

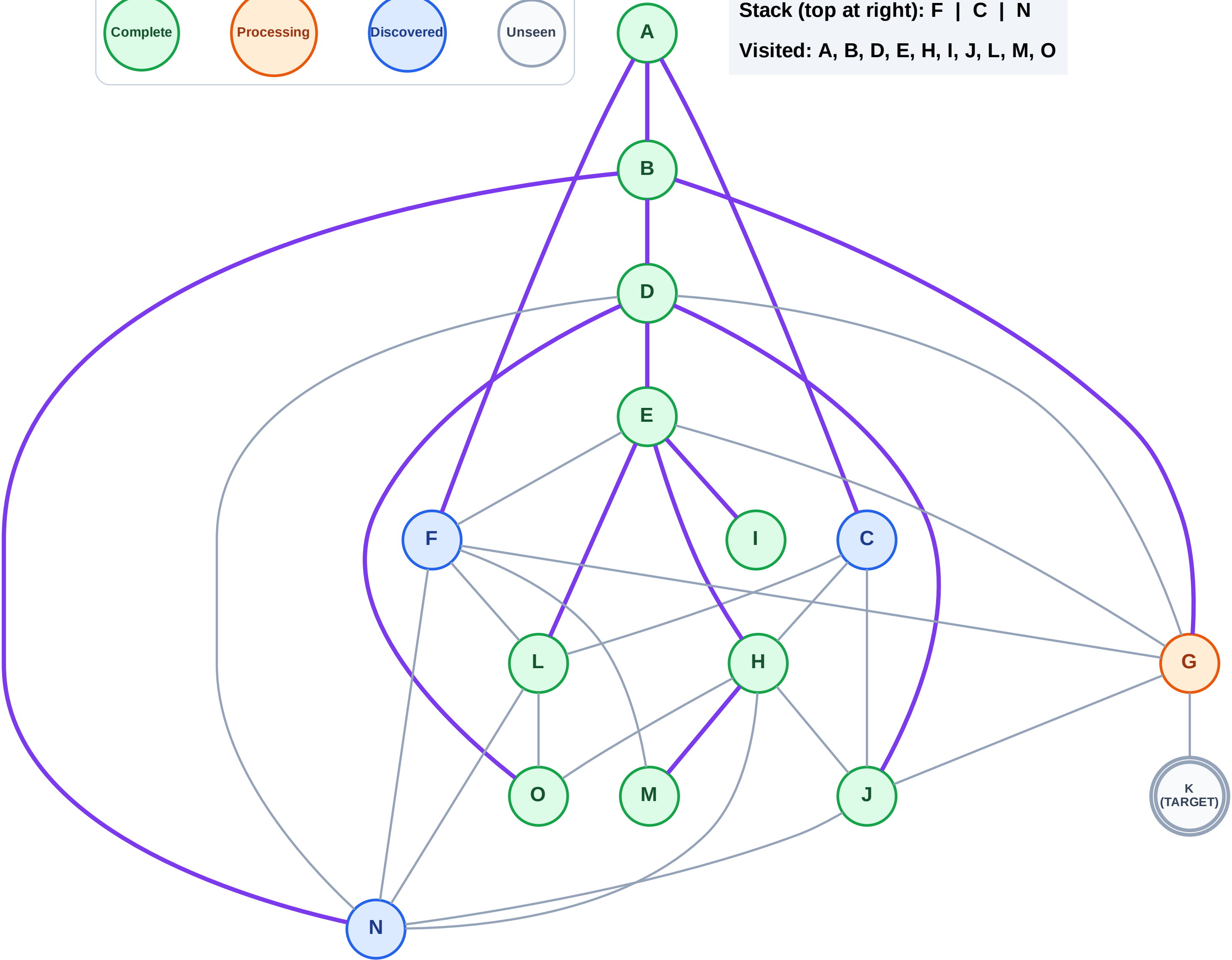
Complete

Processing

Discovered

Unseen

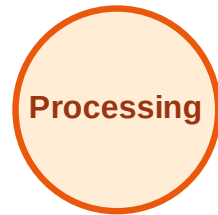
Stack (top at right): F | C | N
Visited: A, B, D, E, H, I, J, L, M, O



DFS Traversal

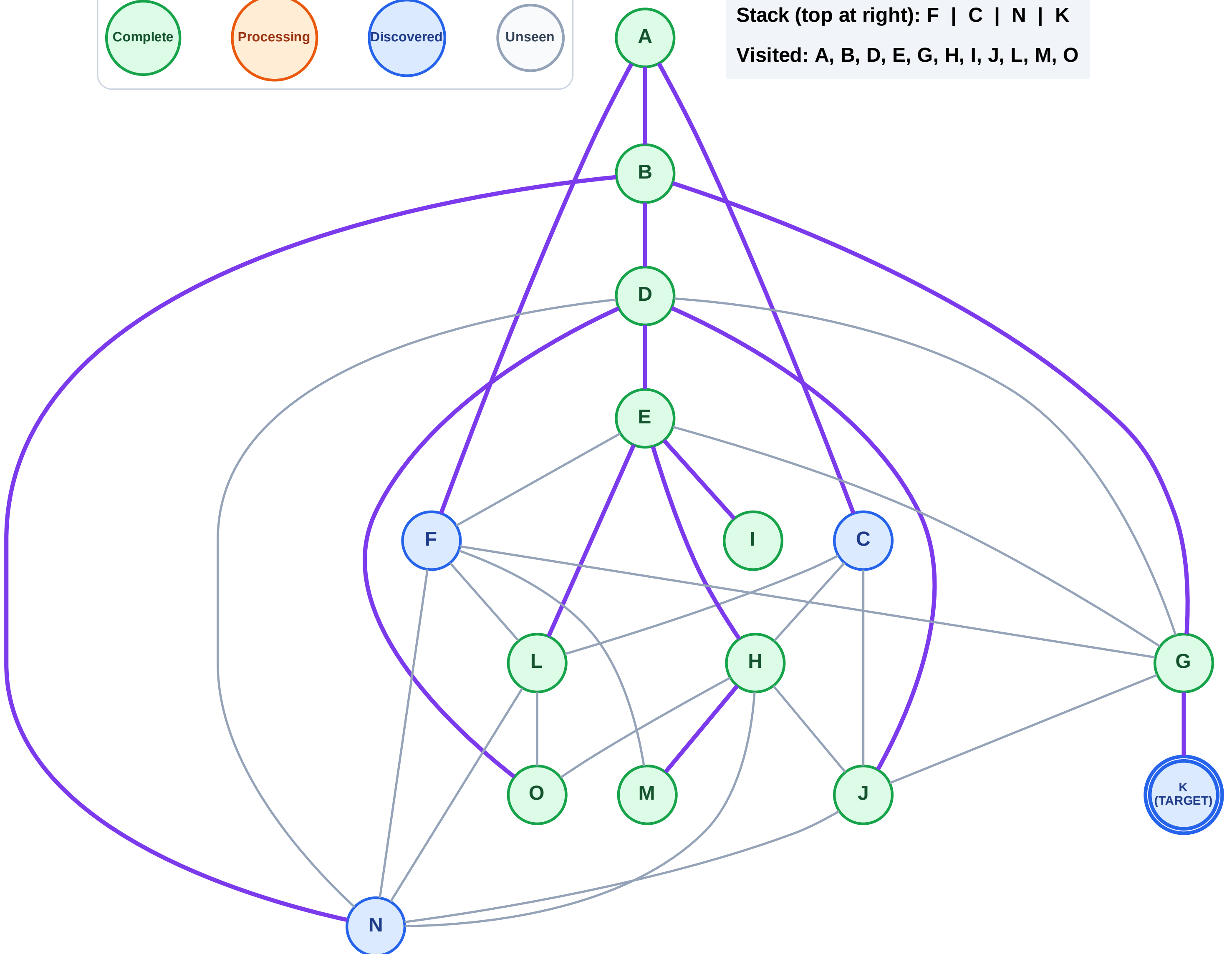
Step 23: Discover K from G

Legend



Stack (top at right): F | C | N | K

Visited: A, B, D, E, G, H, I, J, L, M, O



DFS Traversal

Step 24: Process node K

Legend

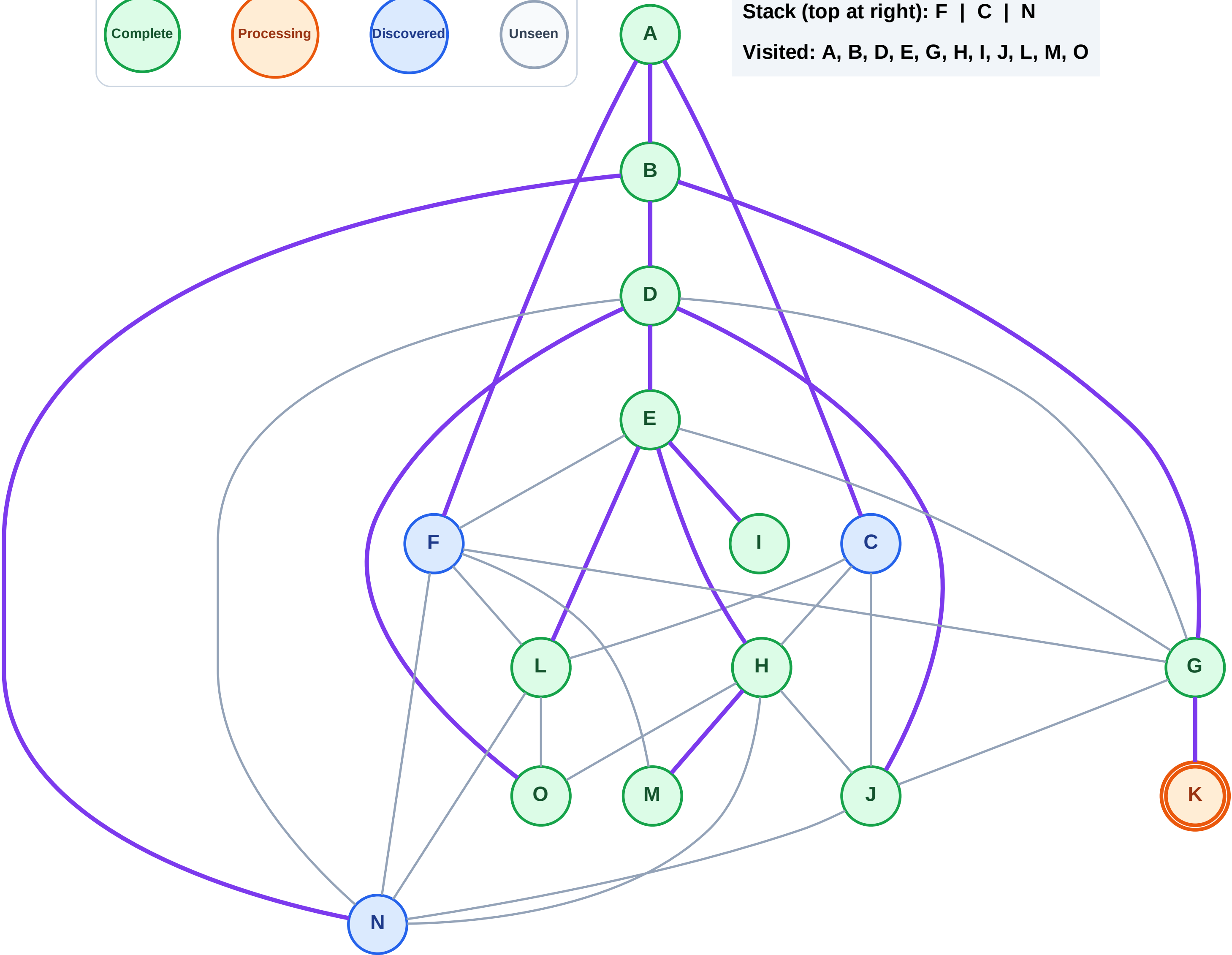
Complete

Processing

Discovered

Unseen

Stack (top at right): F | C | N
Visited: A, B, D, E, G, H, I, J, L, M, O



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (3 hops): A → B → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): F | C | N

Visited: A, B, D, E, G, H, I, J, K, L, M, O

